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ORIGINAL ARTICLE

The implementation and experimental research on an S-curve acceleration and deceleration control algorithm with the characteristics of end-point and target speed modification on the fly

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Abstract For modern equipment manufacturing industries, advanced manufacturing technologies have significant impact on the production quality and efficiency. Specifically, high-speed and high-accuracy technology is widely used in aerospace, automotive, and power-generation equipment to reduce the processing costs and improve the machining accuracy. However, the traditional velocity planning methods cannot satisfy the requirements of advanced equipment in point-to-point movement occasions. To solve these problems, a novel S-curve acceleration and deceleration (Acc/Dec) control model is proposed in this study, and the discretization of the Acc/Dec process was investigated. Furthermore, the derivation of the actual interpolation periods of each segment, actual achieved Acc/Dec, and actual achieved jerk of Acc/Dec zone are addressed in detail. To ensure the positioning accuracy and reduce the computational load, discrete formulae for the deceleration displacement, maximal achieved speed, and displacement of the constant acceleration segment are derived. To satisfy the product diversification and improve the efficiency, the function of the aforementioned S-curve Acc/Dec control algorithm was extended in this study. As a result, the end-point modification and target speed modification algorithms were developed. Finally, a series of numerical simulations and a real experiment were conducted. The results show that the proposed algorithms exhibit a good performance both in terms of reliability and efficiency. Thus, its feasibility was validated.

Keywords S-curve acceleration and deceleration · Point-to-point velocity planning · High-speed and high-accuracy · End-point modification algorithm · Target speed modification algorithm

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Nomenclature

A	The allowable acceleration
A^{real}	The actual achieved acceleration
A_{cur}	The current acceleration/deceleration
D	The allowable deceleration
D^{real}	The actual achieved deceleration
f_i	The feedrate reached at the end of the ith segment, $i = 1, 2, \ldots, 7$
f_s	The starting speed
f_e	The ending speed
F	The commanded feedrate
J_i	The magnitude of jerk in the ith segment, $i = 1, 2, \ldots, 7$
J_{acc}	The desired jerk of the acceleration zone
J_{dec}	The desired jerk of the deceleration zone
$J_{\mathrm{acc}}^{\mathrm{real}}$	The actual achieved jerk of the acceleration zone
$J_{\mathrm{dec}}^{\mathrm{real}}$	The actual achieved jerk of the deceleration zone

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L	The displacement of the programmed path
L_{cur}	The current displacement of tool path
L_{dec}	The deceleration displacement
L_i	The displacement traveled during the ith segment, $i = 1, 2, \ldots, 7$
L_{new}	The new end-point
L_{remain}	The remaining length of the programmed path
M_{state}	The current motion state of Acc/Dec process
n_i	The number of interpolation periods of the ith segment, $i = 1, 2, \ldots, 7$
n_i^{real}	The actual interpolation periods of the ith segment, $i = 1, 2, \ldots, 7$
s_i	The displacement reached at the end of the ith segment, $i = 1, 2, \ldots, 7$
$S_{\mathrm{dec}}(\cdot)$	The deceleration displacement of the speed varied from V_{cur} to f_{e}
V_{cur}	The current speed
$\Delta V_{a\max}$	The maximum allowable speed increment in the acceleration segment
$\Delta V_{d\max}$	The maximum allowable speed increment in the deceleration segment
V_j	The feedrate of the jth interpolation period, $j = 1, 2, \ldots, \sum_{i=1}^{7} n_i$
V_{new}	The new target speed
V_m	The actual maximum feedrate achieved at the end of the decelerated acceleration segment

1 Introduction

With the rapid development of modern equipment manufacturing technologies, high-speed and high-accuracy machining technologies that are important to guarantee machining precision and production efficiency are increasingly demanded in modern manufacturing industries [1, 2]. At present, the modern equipment manufacturing industry has many severe challenges such as increasingly fierce market competition, new requirements to satisfy customer demand, and product diversification [3]. The traditional velocity planning methods cannot satisfy the requirements of modern equipment manufacturing in terms of production efficiency and quality. Particularly, the requirements for advanced equipment applications are higher. For example, the positioning accuracy and efficiency of die bonder equipment are two critical issues that should be solved [4]. However, the positioning accuracy is limited by the vibration of the equipment. Moreover, in response to the decreasing trend of chip thickness, the contact force must be controlled within an appropriate range [5]. Therefore, to improve the productivity and reduce the cost and execution time, it is necessary to develop a new velocity planning method.

To solve the challenges in vibration, the development of a flexible acceleration and deceleration (Acc/Dec) control algorithm is one of the solutions. The performance of the Acc/Dec control algorithm will significantly affect that of the control system. Particularly, real-time and computation load also have a significant impact on the control system [7]. Therefore, an excellent Acc/Dec control algorithm would effectively avoid the vibration, impact, and out-of-step phenomenon occurring during the movement of machine tools. To achieve the smoothness of the feedrate and enhance the positioning accuracy and efficiency, it is very important to develop an efficient Acc/Dec control algorithm. Diverse Acc/Dec control algorithms have been proposed in previous studies. Linear Acc/Dec [8, 9] is the simplest and easiest control algorithm. Nevertheless, its speed profile is not smooth at the end of the acceleration process and at the beginning of the deceleration process. Moreover, its acceleration profile is discontinuous as well. These properties induce the vibration of machine tools during the movement [18]. To improve the movement smoothness, an exponential Acc/Dec control algorithm has been developed [10, 11]. However, the exponential Acc/Dec is too complex and time-consuming to be used in practice. Similarly, its acceleration profile is also not continuous at the beginning and the end of Acc/Dec process. Therefore, a flexible impact on machine tools caused by acceleration gust still exists. To overcome these disadvantages, Wang et al. [12] presented a trigonometric velocity scheduling algorithm for generating continuous velocity, acceleration, and jerk curves. However, the maximum acceleration and jerk cannot be reached during the Acc/Dec process to ensure the velocity profile, causing a conservative velocity planning. Besides, it is also more complex and needs extensive computation, thus failing to satisfy the real-time requirement of control systems. To further reduce the vibration and impact on machine tools, higher-order Acc/Dec control algorithms have been studied. Song et al. [7] proposed a novel fourth-order S-shape Acc/Dec control algorithm by the use of the parameter restriction that applied in CNC system. Similarly, Leng et al. [13] developed a cubic polynomial Acc/Dec control model. In general, the computation load significantly increases with the increase in the polynomial order. Therefore, higher-degree polynomial profiles demand more computational efforts. What is worse, the computational complexity increases almost exponentially along with the order of polynomial.

To decrease the vibration of machine tools during the machining, Erkorkmaz et al. [6] first presented the S-curve Acc/Dec control algorithm in 2001. Compared with the linear and exponential Acc/Dec control algorithms, it generates smoother commands, resulting in a better accuracy. Moreover, it requires less computation load and storage space than a high-order Acc/Dec control algorithm. Therefore, the S-curve Acc/Dec control algorithm has attracted much attention. As a result, an adaptive S-curve Acc/Dec control method was proposed by Zheng et al. [14]. Later, He et al. [15] designed a velocity planning model based on the jerk symmetry in S-shape Acc/Dec control algorithm. To improve the machinery dynamics by reducing the jerk value, Chen et al. [16] employed a smooth Scurve profiling. To reduce the computation time, Xia et al. [17] used an S-curve Acc/Dec based on reference table which may limit the flexibility of the algorithm. Similarly, Wang et al. [18] reported a novel look-ahead and adaptive speed control algorithm based on preinterpolation S-curve Acc/Dec control algorithm to satisfy the requirements for high-speed machining. Subsequently, Chen et al. [19] reported a feedrate profile-generation approach and the corresponding look-ahead algorithm with whole S-curve Acc/Dec control algorithm. However, these methods assume that the velocity curve is symmetric to simplify the planning process. This restricts the dynamic motion of the axes. Moreover, additional restrictions (such as symmetric S-shape) would increase the solution space and achieve the optimal solutions easily. To increase the planning efficiency, Lu et al. [20] designed a genetic algorithm-based S-curve Acc/Dec scheme, yet used a simplified five-segment S-curve Acc/Dec control algorithm which probably causes motor overload when machine tools move with high speed. Later, Qiao et al. [21] reported a four-level trajectory planning strategy including position, velocity, acceleration, and jerk planning. Dong et al. [22] employed a seven-segment jerk-limited S-curve-based feedrate profile planning for continuous short line tool path with contour error constraint. Fan et al. and others [23, 24] proposed a 15-phase Acc/Dec mode with confined jounce, jerk, acceleration, and speed for parametric tool paths. Additional methods for generating interpolation parameters and higher-order Acc/Dec will significantly increase the complexity of the algorithm and computation cost. Moreover, to the best of our knowledge, the end-point and target speed modification algorithms have been rarely investigated. Nevertheless, these functions would be very important for high-speed and high-accuracy positioning applications, such as dispenser machine, die bonder machine, and SMT machine. Utilization of these algorithms not only extends the functions of a control system and satisfies diverse customer requirements but also saves the execution time and improves the efficiency.

To overcome the abovementioned shortcomings, we first constructed an unsymmetrical S-curve Acc/Dec control algorithm with confined jerk, acceleration, and speed for high-speed and high-accuracy point-to-point (PTP) movement occasions. Then, the function of the S-curve Acc/Dec control algorithm was extended to increase the throughput of equipment, including the end-point modification algorithm (EPMA) and target speed modification algorithm (TSMA). The remainder of this paper is organized as follows. A brief introduction on the kinematic profiles and the basic formulae of the S-curve Acc/Dec control algorithm are given in Sect. 2. The velocity planning of the S-curve Acc/Dec control algorithm, including the discretization of Acc/Dec zone, computation of maximal achieved speed, deceleration point prediction, remaining displacement compensation, and implementation of velocity planning process are described in Sect. 3. The extended functionality of the S-curve Acc/Dec control algorithm including EPMA and TSMA is described in Sect. 4. Algorithm simulation and experimental results are discussed in Sect. 5. Conclusions are summarized in Sect. 6.

2 Principle of S-curve Acc/Dec control algorithm

In this section, the kinematic profiles and formulae are introduced.

2.1 Kinematic profiles

The kinematic profiles of seven-segment S-curve Acc/Dec process are shown in Fig. 1. In general, it consists of an acceleration zone, a constant speed zone, and a deceleration zone. As shown in Fig. 1 (a), the displacement profile can be subdivided into seven segments from I to VII, namely, accelerated acceleration segment I, constant acceleration segment II, decelerated acceleration segment III, constant speed segment IV,

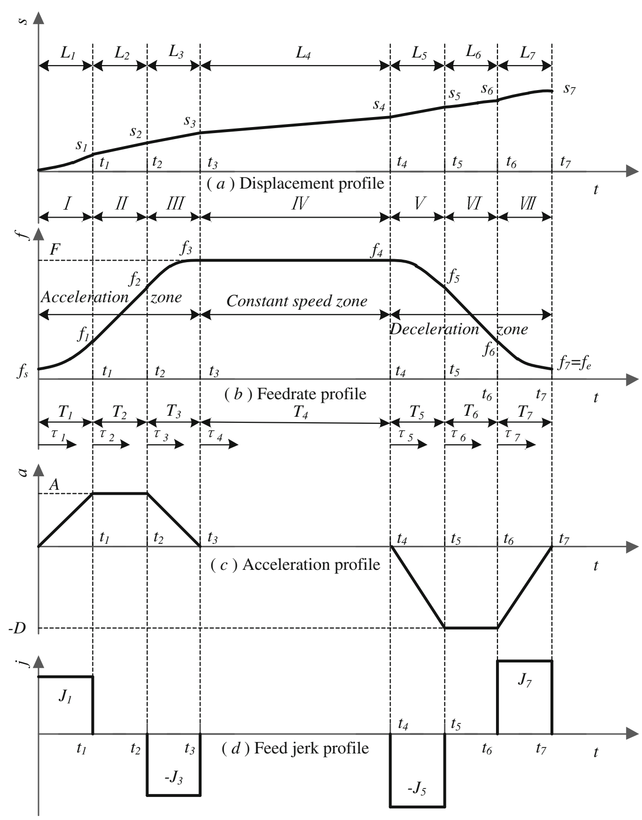

Fig. 1 Kinematic profiles of seven-segment S-curve Acc/Dec process

accelerated deceleration segment V, constant deceleration segment VI, and decelerated deceleration segment VII [6, 14].

The displacement of the programmed path, allowable A and D, iteration time of interpolation, and commanded feedrate were considered when generating a speed profile. When the constraint conditions changed, segment II, IV, or VI may not exist.

2.2 Formulae

The input parameters include the displacement of the programmed path (L), starting speed (f_s), commanded feedrate (F), ending speed (f_e), allowable acceleration (A), allowable deceleration (D), desired jerk of the acceleration zone (J_{acc}), desired jerk of the deceleration zone (J_{dec}), and sampling period (T_s). To construct an unsymmetrical S-curve Acc/Dec control algorithm, asymmetric A/D as well as asymmetrical $J_{\mathrm{acc}}/J_{\mathrm{dec}}$ are incorporated by the user. Furthermore, for a smooth connection of the adjacent segment, nonzero f_s and f_e are allowed. Suppose the initial conditions of jerk, acceleration, feedrate, and displacement at t_i (i = 1, 2, … , 7) are known, the relationship among acceleration $a(t)$, feedrate $f(t)$, and displacement $s(t)$ profiles can be expressed as follows:

$$\begin{cases} a(t) = a(t_i) + \int_{t_i}^{t} j(\tau_i)\mathrm{d}\tau_i \\ f(t) = f(t_i) + \int_{t_i}^{t} a(\tau_i)\mathrm{d}\tau_i \\ s(t) = s(t_i) + \int_{t_i}^{t} f(\tau_i)\mathrm{d}\tau_i \end{cases} \tag{1}$$

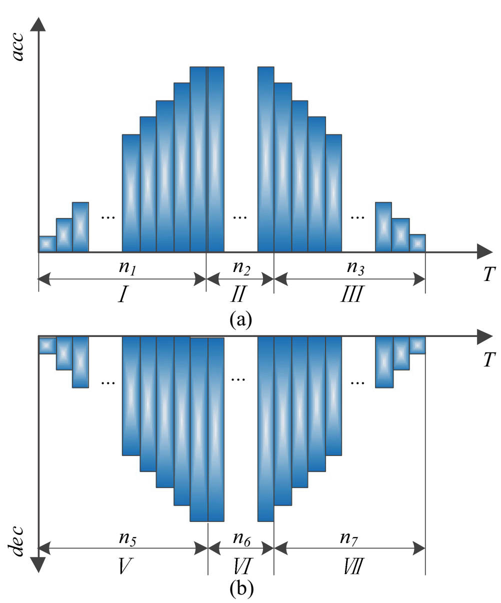

Fig. 2 The discretization of the Acc/Dec zone

The feed jerk profile shown in Fig. 1 (d) can be expressed as follows:

$$j(\tau_i) = \begin{cases} J_1, \tau_1 \in [0, t_1) \\ 0, \tau_2 \in [t_1, t_2) \\ -J_3, \tau_3 \in [t_2, t_3) \\ 0, \tau_4 \in [t_3, t_4) \\ -J_5, \tau_5 \in [t_4, t_5) \\ 0, \tau_6 \in [t_5, t_6) \\ J_7, \tau_7 \in [t_6, t_7) \end{cases} \tag{2}$$

where t denotes the absolute time, $t_i (i = 1, 2, \ldots, 7)$ is the time boundary of each segment. By integrating Eq. (2) with respect to time, the acceleration profile shown in Fig. 1(c) can be obtained as follows:

$$a(\tau_i) = \begin{cases} J_1\tau_1, \tau_1 \in [0, t_1) \\ A, \tau_2 \in [t_1, t_2) \\ A - J_3\tau_3, \tau_3 \in [t_2, t_3) \\ 0, \tau_4 \in [t_3, t_4) \\ -J_5\tau_5, \tau_5 \in [t_4, t_5) \\ -D, \tau_6 \in [t_5, t_6) \\ -D + J_7\tau_7, \tau_7 \in [t_6, t_7) \end{cases} \tag{3}$$

where $\tau_i = t - t_{i-1}$ $(i = 1, 2, \ldots, 7)$ is the relative time parameter that starts at the beginning of the ith segment. Similarly, the feedrate profile shown in Fig. 1 (b) can be expressed as follows:

$$f(\tau_i) = \begin{cases} f_s + \frac{1}{2}J_1\tau_1^2, \tau_1 \in [0, t_1) \\ f_1 + A\tau_2, \tau_2 \in [t_1, t_2), \ \text{where} \ f_1 = f_s + \frac{1}{2}J_1T_1^2 \\ f_2 + A\tau_3 - \frac{1}{2}J_3\tau_3^2, \tau_3 \in [t_2, t_3), \ \text{where} \ f_2 = f_1 + AT_2 \\ f_3, \tau_4 \in [t_3, t_4), \ \text{where} \ f_3 = f_2 + AT_3 - \frac{1}{2}J_3T_3^2 = F \\ f_4 - \frac{1}{2}J_5\tau_5^2, \tau_5 \in [t_4, t_5), \ \text{where} \ f_4 = f_3 \\ f_5 - D\tau_6, \tau_6 \in [t_5, t_6), \ \text{wherre} \ f_5 = f_4 - \frac{1}{2}J_5T_5^2 \\ f_6 - D\tau_7 + \frac{1}{2}J_7\tau_7^2, \tau_7 \in [t_6, t_7), \ \text{where} \ f_6 = f_5 - DT_6 \\ f_7 = f_e \end{cases} \tag{4}$$

where T_i (i = 1, 2,…, 7) is the duration of the ith segment. By integrating Eq. (4) with respect to time, the displacement profile shown in Fig. 1 (a) is obtained as follows:

$$s(\tau_i) = \begin{cases} S_0 + f_s\tau_1 + \frac{1}{6}J_1\tau_1^3, \tau_1 \in [0, t_1), \ \text{where} \ s_0 = 0 \\ S_1 + f_1\tau_2 + \frac{1}{2}A\tau_2^2, \tau_2 \in [t_1, t_2), \ \text{where} \ s_1 = s_0 + f_sT_1 + \frac{1}{6}J_1T_1^3 \\ S_2 + f_2\tau_3 + \frac{1}{2}A\tau_3^2 - \frac{1}{6}J_3\tau_3^3, \tau_3 \in [t_2, t_3), \ \text{where} \ s_2 = s_1 + f_1T_2 + \frac{1}{2}AT_2^2 \\ S_3 + f_3\tau_4, \tau_4 \in [t_3, t_4), \ \text{where} \ s_3 = s_2 + f_2T_3 + \frac{1}{2}AT_3^2 - \frac{1}{6}J_3T_3^3 \\ S_4 + f_4\tau_5 - \frac{1}{6}J_5\tau_5^3, \tau_5 \in [t_4, t_5), \ \text{where} \ s_4 = s_3 + f_3T_4 \\ S_5 + f_5\tau_6 - \frac{1}{2}D\tau_6^2, \tau_6 \in [t_5, t_6), \ \text{where} \ s_5 = s_4 + f_4T_5 - \frac{1}{6}J_5T_5^3 \\ S_6 + f_6\tau_7 - \frac{1}{2}D\tau_7^2 + \frac{1}{6}J_7\tau_7^3, \tau_7 \in [t_6, t_7), \ \text{where} \ s_6 = s_5 + f_5T_6 - \frac{1}{2}DT_6^2 \\ S_7 = L \end{cases} \tag{5}$$

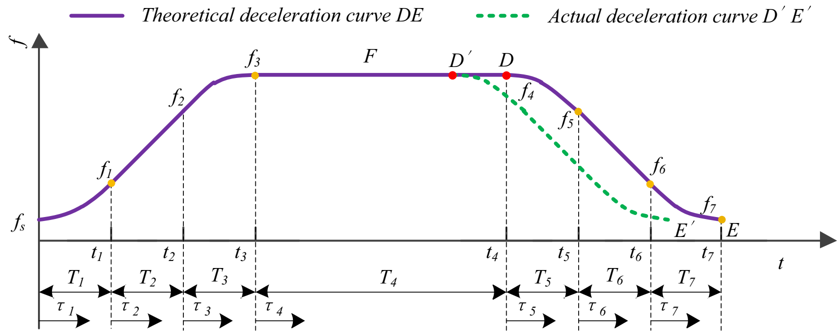

Fig. 3 Deceleration point prediction scheme

For the convenience of calculation, the displacement traveled during each segment can be derived as follows:

$$L_i=\begin{cases} L_1=s_1=s_0+f_sT_1+\dfrac{1}{6}J_1T_1^3, s_0=0\\ L_2=s_2-s_1=f_1T_2+\dfrac{1}{2}A\ T_2^2\\ L_3=s_3-s_2=f_2T_3+\dfrac{1}{2}AT_3^2-\dfrac{1}{6}J_3T_3^3\\ L_4=s_4-s_3=f_3T_4\\ L_5=s_5-s_4=f_4T_5-\dfrac{1}{6}J_5T_5^3\\ L_6=s_6-s_5=f_5T_6-\dfrac{1}{2}D\ T_6^2\\ L_7=s_7-s_6=f_6T_7-\dfrac{1}{2}DT_7^2+\dfrac{1}{6}J_7T_7^3 \end{cases} \tag{6}$$

3 Velocity planning of the S-curve Acc/Dec control algorithm

Because most of the NC system is based on a data-sampling control system, the process time of each segment should be integral times of the sampling period. However, errors often occur because of the limitation of processor and other factors. Therefore, a discretization error was introduced here and defined as a rounding error. To simplify the calculation, assume the interpolation periods of acceleration (deceleration) varied from 0 to maximum equal to those of acceleration (deceleration) varied from maximum to 0; thus,

$$\begin{cases} J_1=J_3=J_{\mathrm{acc}}\\ J_5=J_7=J_{\mathrm{dec}}\\ T_1=T_3=\dfrac{A}{J_1}=\dfrac{A}{J_3}\\ T_5=T_7=\dfrac{D}{J_5}=\dfrac{D}{J_7} \end{cases} \tag{7}$$

Next, the discrete S-curve speed control algorithm was derived. It consists of the discretization of the acceleration (deceleration) zone, maximal achieved speed computation, deceleration point prediction, remaining displacement compensation, and velocity planning process implementation.

3.1 Discretization of acceleration zone

Figure 2a, b shows the discretization of the Acc/Dec zone, respectively. Figure 2a shows the relationship between n_1 and n_3 as follows:

$$n_3=n_1-1 \tag{8}$$

By rounding n_1 upwards to the nearest integer according to Eq. (7), we obtained

$$n_1=\mathrm{ceil}\left(\frac{A}{J_{\mathrm{acc}}}\right) \tag{9}$$

where ceil (x) indicates that x is rounded upwards to the nearest integer. Similarly, the total speed increment of segments I and III can be calculated as follows:

$$\Delta V_{\mathrm{acc}}=\sum\nolimits_{i=1}^{n_1}(i\times J_{\mathrm{acc}})+\sum\nolimits_{i=1}^{n_3}(A-i\times J_{\mathrm{acc}}) \tag{10}$$

By simplifying, Eq. (10) can be rewritten as follows:

$$\Delta V_{\mathrm{acc}}=n_1\times A=n_1^2\times J_{\mathrm{acc}} \tag{11}$$

The maximum allowable speed increment can be derived as follows:

$$\Delta V_{\mathrm{amax}}=F-f_s \tag{12}$$

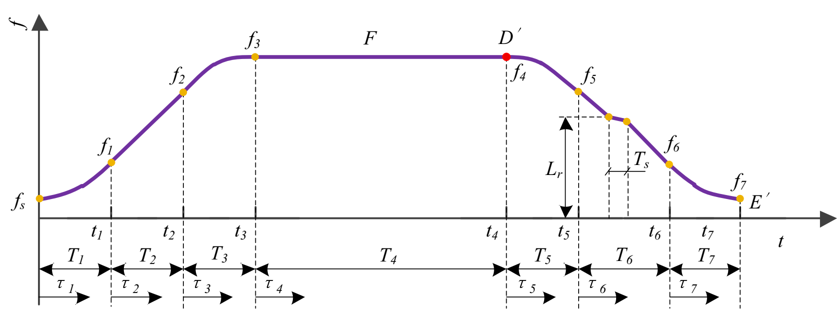

Fig. 4 Schematic of the remaining displacement compensation

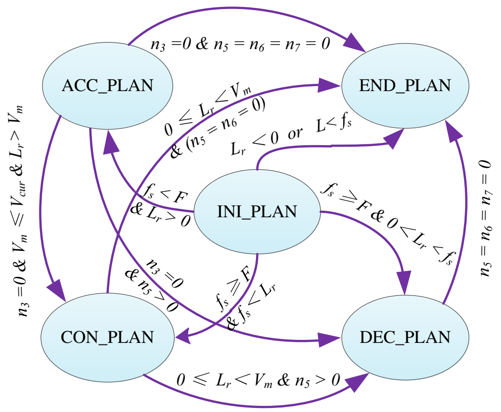

Fig. 5 State transition diagram of FSM

According to whether there is a segment II, the acceleration profile can be divided into two cases:

3.1.1 Case a: $\Delta V_{acc} \geq \Delta V_{amax}$

The allowable acceleration A is not reachable, namely, $n_2 = 0$. Considering Eqs. (11) and (12), n_1^{real} can be calculated as follows:

$$n_1^{\text{real}} = \text{ceil}\left(\sqrt{\frac{\Delta V_{amax}}{J_{\text{acc}}}}\right) \tag{13}$$

Then, $J_{\text{acc}}^{\text{real}}$ and A^{real} can be recalculated as follows:

$$J_{\text{acc}}^{\text{real}} = \frac{\Delta V_{amax}}{n_1^{\text{real}} \times n_1^{\text{real}}}, \quad A^{\text{real}} = n_1^{\text{real}} \times J_{\text{acc}}^{\text{real}} \tag{14}$$

Therefore, n_2^{real} and n_3^{real} can be expressed as follows:

$$n_2^{\text{real}} = 0, \quad n_3^{\text{real}} = n_1^{\text{real}} - 1 \tag{15}$$

3.1.2 Case B: $\Delta V_{acc} < \Delta V_{amax}$

In this situation, A is reachable, namely, $n_2 > 0$. The total speed increment of segment II is derived as $n_2 \times A$. Combined with Eq. (11), ΔV_{amax} can be calculated as follows:

$$\Delta V_{amax} = n_a \times A \tag{16}$$

where $n_a = n_1 + n_2$; then, n_a^{real} is obtained as follows:

$$n_a^{\text{real}} = \text{ceil}\left(\frac{\Delta V_{amax}}{A}\right) \tag{17}$$

Therefore, A^{real}, n_1^{real} and $J_{\text{acc}}^{\text{real}}$ were readjusted as follows:

$$A^{\text{real}} = \frac{\Delta V_{amax}}{n_a^{\text{real}}} = \frac{\Delta V_{amax}}{(n_1 + n_2)}, \quad n_1^{\text{real}} = \text{ceil}\left(\frac{A^{\text{real}}}{J_{\text{acc}}}\right), \quad J_{\text{acc}}^{\text{real}} = \frac{A^{\text{real}}}{n_1^{\text{real}}} \tag{18}$$

Then, n_2^{real} and n_3^{real} can be obtained as follows:

$$n_2^{\text{real}} = n_a^{\text{real}} - n_1^{\text{real}}, \; n_3^{\text{real}} = n_1^{\text{real}} - 1 \tag{19}$$

3.2 Discretization of deceleration zone

Figure 2b shows the discretization of the deceleration zone. Similarly, n_5 and n_7 are related as follows:

$$n_7 = n_5 - 1 \tag{20}$$

Similarly, n_5 can be solved as follows:

$$n_5 = \text{ceil}\left(\frac{D}{J_{\text{dec}}}\right) \tag{21}$$

Then, the total speed increment of segments V and VII can be expressed as follows:

$$\Delta V_{\text{dec}} = \left|\sum\nolimits_{i=1}^{n_5} (-i \times J_{\text{dec}})\right| + \left|\sum\nolimits_{i=1}^{n_7} (-D + i \times J_{\text{dec}})\right| \tag{22}$$

Table 1 Definition and transition conditions of states

Present state	Definition	Triggering conditions	Next state
INI_PLAN	Start state	$f_s < F$ and $L_r > 0$	ACC_PLAN
		$f_s \geq F$ and $f_s < L_r$	CON_PLAN
		$f_s \geq F$ and $0 < L_r < f_s$	DEC_PLAN
		$L_r < 0 \| L < f_s$	END_PLAN
ACC_PLAN	Acceleration state	$n_3 = 0$ and $V_m \leq V_{cur}$ and $L_r > V_m$	CON_PLAN
		$n_3 = 0$ and $n_5 > 0$	DEC_PLAN
		$n_3 = 0$ and $n_5 = n_6 = n_7 = 0$	END_PLAN
CON_PLAN	Constant speed state	$0 \leq L_r < V_m$ and $(n_5 > 0)$	DEC_PLAN
		$0 \leq L_r < V_m$ and $(n_5 = n_6 = 0)$	END_PLAN
DEC_PLAN	Deceleration state	$n_5 = n_6 = n_7 = 0$	END_PLAN
END_PLAN	End state	–	/

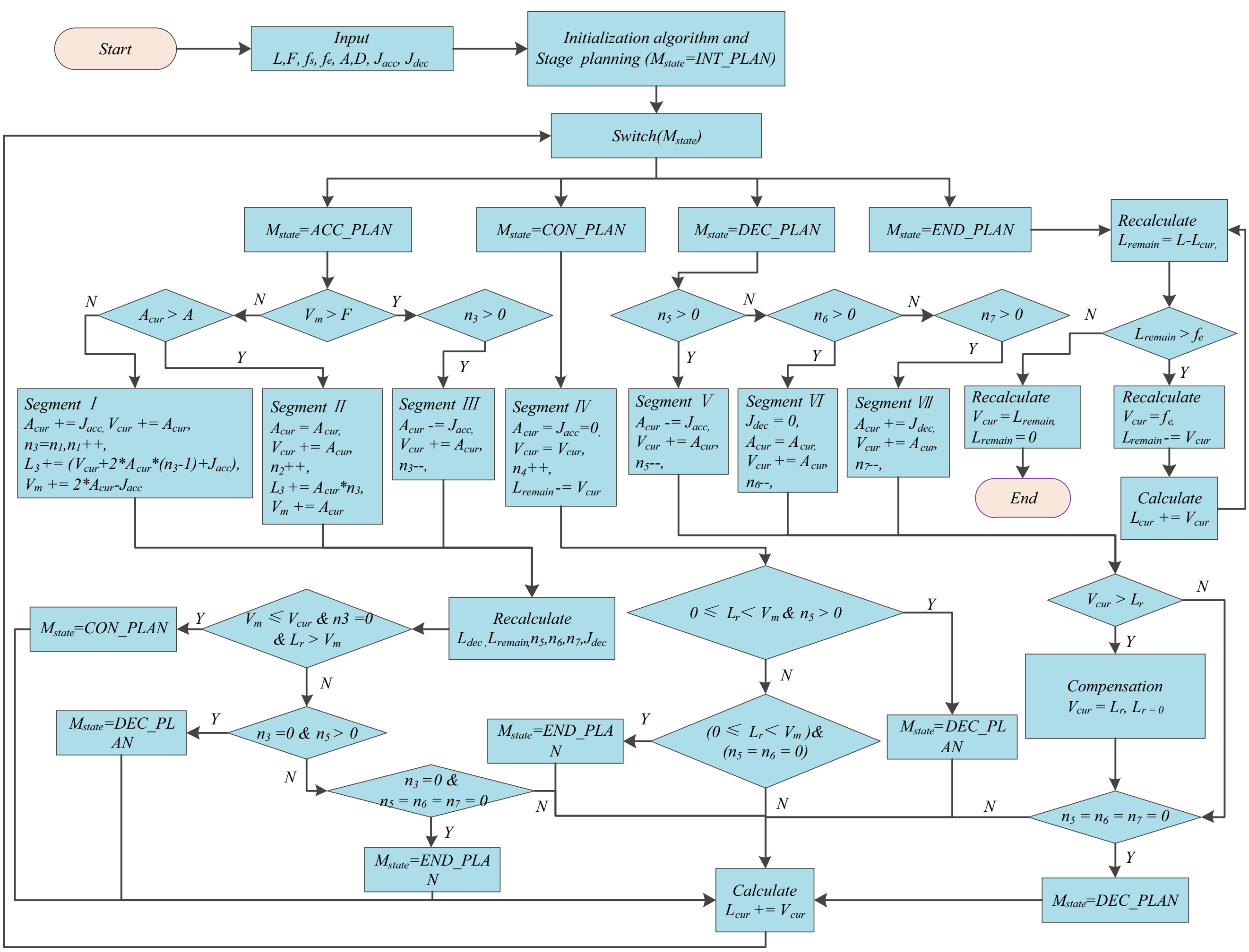

Fig. 6 Flowchart of the S-curve Acc/Dec velocity planning process

where |·| indicates the absolute value. By simplifying, Eq. (22) can be rewritten as follows:

$$\Delta V_{\mathrm{dec}} = n_5 \times D = n_5^2 \times J_{\mathrm{dec}} \tag{23}$$

The maximum allowable speed increment can be derived as follows:

$$\Delta V_{d\max} = F - f_e \tag{24}$$

According to whether there is a segment VI, the deceleration profile can be divided into two cases:

3.2.1 Case a: $\Delta V_{dec} \geq \Delta V_{dmax}$

In this case, segment VI is absent. According to Eqs. (23) and (24), n_5^{real} can be solved as follows:

$$n_5^{\mathrm{real}} = \mathrm{ceil}\left(\sqrt{\frac{\Delta V_{d\max}}{J_{\mathrm{dec}}}}\right) \tag{25}$$

Similarly, $J_{\mathrm{dec}}^{\mathrm{real}}$ and D^{real} can be expressed as follows:

$$J_{\mathrm{dec}}^{\mathrm{real}} = \frac{\Delta V_{d\max}}{n_5^{\mathrm{real}} \times n_5^{\mathrm{real}}}, \quad D^{\mathrm{real}} = n_5^{\mathrm{real}} \times J_{\mathrm{dec}}^{\mathrm{real}} \tag{26}$$

Then, n_6^{real} and n_7^{real} can be computed as follows:

$$n_6^{\mathrm{real}} = 0, \quad n_7^{\mathrm{real}} = n_5^{\mathrm{real}} - 1 \tag{27}$$

3.2.2 Case B: $\Delta V_{dec} < \Delta V_{dmax}$

In the circumstances, D can be achieved, indicating $n_6 > 0$. Then, the speed increment of segment VI is derived as $n_6 \times D$. Combined with Eq. (23), $\Delta V_{d\max}$ can be derived as follows:

$$\Delta V_{d\max} = n_d \times D \tag{28}$$

where $n_d = n_5 + n_6$; therefore, n_d^{real} can be derived as follows:

$$n_d^{\mathrm{real}} = \mathrm{ceil}\left(\frac{\Delta V_{d\max}}{D}\right) \tag{29}$$

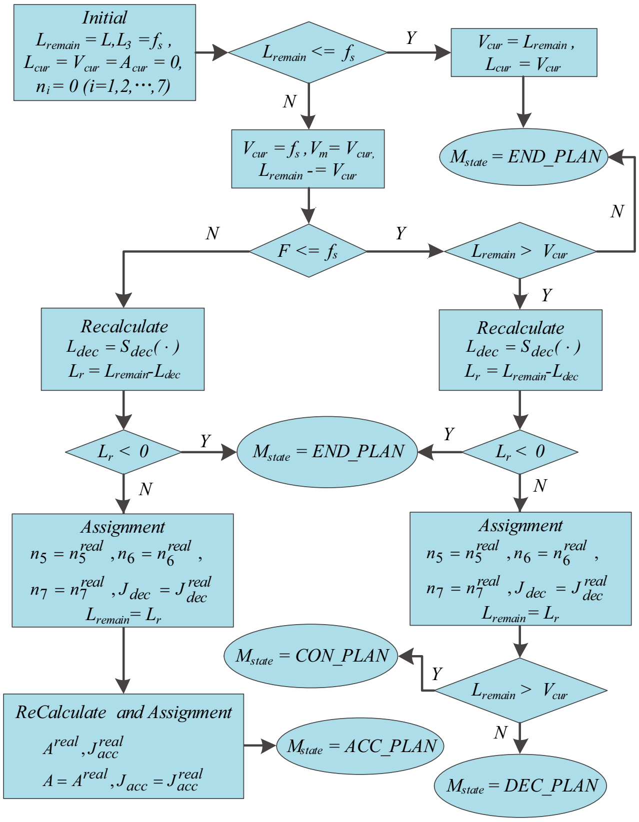

Fig. 7 Flowchart of the initialization algorithm and stage planning

Then, D^{real}, n_5^{real} and $J_{\text{dec}}^{\text{real}}$ can be respectively calculated as follows:

$$D^{\text{real}} = \frac{\Delta V_{d\max}}{n_d^{\text{real}}}, \quad n_5^{\text{real}} = \text{ceil}\left(\frac{D^{\text{real}}}{J_{\text{dec}}}\right), \quad J_{\text{dec}}^{\text{real}} = \frac{D^{\text{real}}}{n_5^{\text{real}}} \quad (30)$$

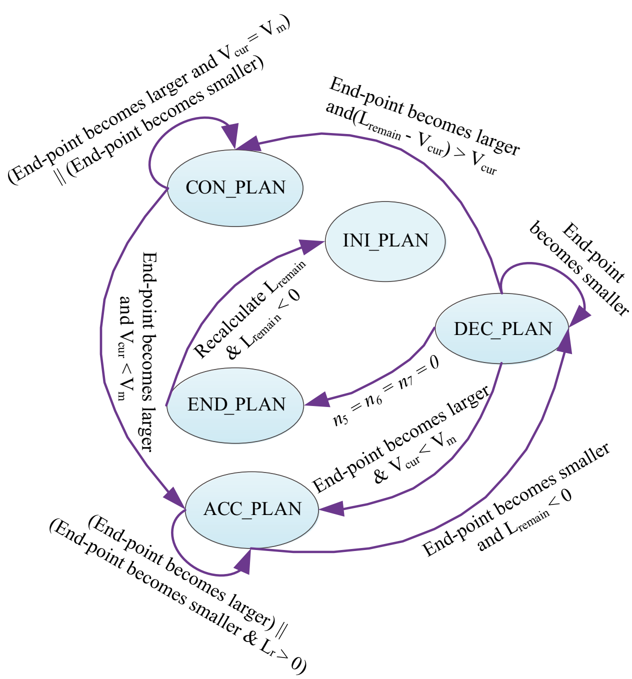

Fig. 8 State transition diagram of EPMA

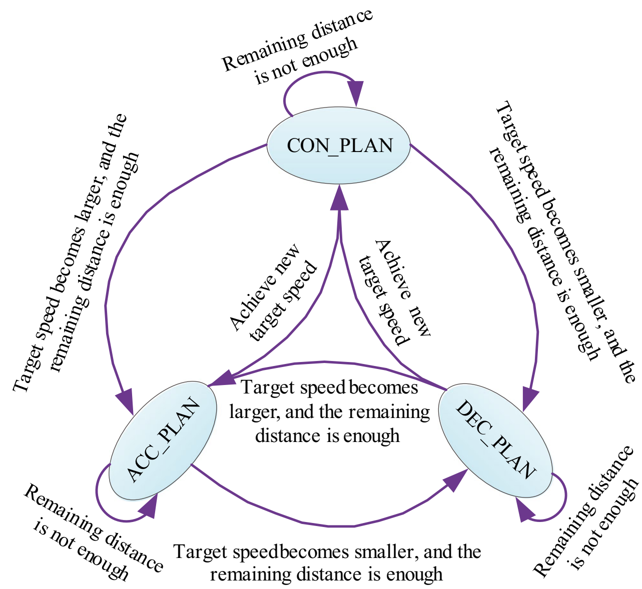

Fig. 9 State transition diagram of TSMA

Thus, n_6^{real} and n_7^{real} can be calculated as follows:

$$n_6^{\text{real}} = n_d^{\text{real}} - n_5^{\text{real}}, n_7^{\text{real}} = n_5^{\text{real}} - 1 \quad (31)$$

3.3 Maximal achieved speed computation

The maximal achieved speed should be satisfied as follows:

$$V_m \leq F \quad (32)$$

where V_m denotes the maximal achieved speed at the end of segment III. Therefore, to satisfy the inequality (32), the speed achieved in the next period should be precalculated in real time during segments I and II, and the Acc/Dec process will become segment III in the next period once Eq. (32) is satisfied.

According to Eq. (3), the acceleration of segment III can be precomputed as follows:

$$a_i = A_{\text{cur}} - i \times J_{\text{acc}}^{\text{real}} \quad (33)$$

where a_i denotes the acceleration achieved in the next period. A_{cur} is the current acceleration. By integrating Eq. (33), the speed of segment III can be precomputed as follows:

$$V_i = V_{\text{cur}} + \Sigma_{i=1}^{n} a_i = V_{\text{cur}} + A_{\text{cur}} \times n - \frac{n \times (n+1)}{2} \times J_{\text{acc}}^{\text{real}} \quad (34)$$

where V_i denotes the speed achieved in the next period. V_{cur} represents the current speed. By substituting n_3 into Eq. (34), V_m can be obtained as follows:

$$V_m = V_{\text{cur}} + \Sigma_{i=1}^{n_3} a_i = V_{\text{cur}} + A_{\text{cur}} \times n_3 - \frac{n_3 \times (n_3+1)}{2} \times J_{\text{acc}}^{\text{real}} \quad (35)$$

Table 2 Input parameters and simulation results of the S-curve Acc/Dec control algorithm

Group	Input parameters							Actual achieved value		
	L (mm)	f_s (mm/s)	f_e (mm/s)	F (mm/s)	$A/D \times 10^3$ (mm/s^2)	$J_{acc}/J_{dec} \times 10^6$ (mm/s^3)	T_s (ms)	V_m (mm/s)	$A/D \times 10^3$ (mm/s^2)	$J_{acc}/J_{dec} \times 10^6$ (mm/s^3)
0	30	1	2	1000	20/30	4/5	0.5	780.2	19.98/29.93	3.996/4.988
1	30	1	2	500	100/100	10/10		500	62.38/62.25	7.796/7.781
2	30	2	2	500	30/30	5/5		500	29.29/29.29	4.882/4.882

The use of Eq. (35) to calculate V_m consumes a lot of time. Therefore, a simplified solution to calculate V_m should be obtained. The derivation of the simplified solution is given in Appendix 1. The following computations directly provide the formula:

(1) In segment I, V_m can be computed as follows:

$$V_m = V'_m + A_{\mathrm{cur}} \times 2 - J^{\mathrm{real}}_{\mathrm{acc}} \tag{36}$$

(2) In segment II, V_m can be computed as follows:

$$V_m = V'_m + A_{\mathrm{cur}} \tag{37}$$

3.4 Deceleration point prediction

To ensure the accuracy of the final position, it is required to compute and compare the deceleration displacement with the remaining length of the programmed path for each sampling period to determine whether it should enter the deceleration zone in the next interpolation period. Figure 3 shows the deceleration point prediction scheme. Without the loss of generality, we suppose that L is long enough to achieve the commanded feedrate F at the end of segment III.

As shown in Fig. 3, curve DE and point D denote the theoretical deceleration curve and deceleration point, respectively. Curve $D'E'$ and point D' denote the actual deceleration curve and deceleration point, respectively. Considering L is not usually an integer multiple of the discrete feedrate, the following inequality

$$0 \le L_{\mathrm{remain}} - L_{\mathrm{dec}} < V_m \tag{38}$$

should be satisfied at the final interpolation period of segment IV. V_m is equal to F if L is long enough. Or, V_m can be

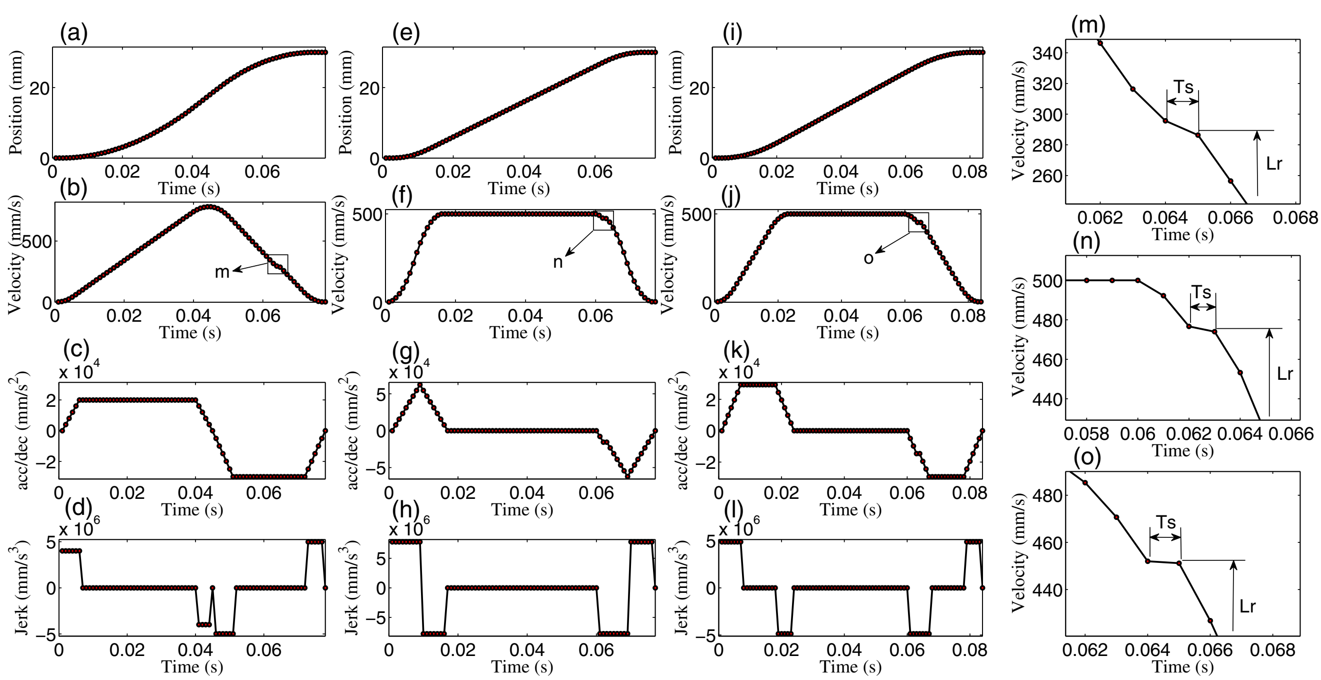

Fig. 10 Simulation on S-curve Acc/Dec control algorithm. **a**, **e**, **i** Displacement profile; **b**, **f**, **j** speed profile; **c**, **g**, **k** acceleration/deceleration profile; **d**, **h**, **l** jerk profile; **m**, **n**, **o** partial enlargement of (**b**), (**f**), and (**j**), respectively

Table 3 Input parameters of end-point modification algorithm

L (mm)	f_s (mm/s)	f_e (mm/s)	F (mm/s)	$A/D \times 10^3$ (mm/s^2)	$J_{acc}/J_{dec} \times 10^6$ (mm/s^3)
30	0	0	1000	100/100	10/10

calculated using Eq. (36) or (37). Assuming that the Acc/Dec process at the final interpolation period of segment IV is located at point D' and $L_{D'D} = L_{\text{remain}} - L_{\text{dec}}$, Eq. (38) can be rewritten as follows:

$$0 \le L_{D'D} < V_m \tag{39}$$

where the equal sign is formed if and only if the theoretical deceleration point D coincides with the actual deceleration point D'. For convenience, the Acc/Dec process will become the deceleration zone according to the deceleration curve $D'E'$ in advance, and the remaining displacement $L_{D'D}$ will be compensated during the deceleration zone. To determine point D', the deceleration displacement L_{dec} should be calculated in real time. Generally, L_{dec} consists of L_5, L_6, and L_7. Nevertheless, once the segment I exists, L_{dec} also includes L_3 to ensure the continuity of acceleration. The calculation of L_{dec} is derived as follows:

3.4.1 Displacement of deceleration zone

(1) Compute L_5

According to Eqs. (3), (4), and (6), the deceleration, feedrate, and displacement of segment V can be computed as follows:

$$\begin{cases} a_i = -J_{\text{dec}}^{\text{real}} \times i \\ f_i = V_m - J_{\text{dec}}^{\text{real}} \times \dfrac{i \times (i+1)}{2} \\ L_i = V_m \times i - J_{\text{dec}}^{\text{real}} \times \dfrac{i \times (i+1) \times (i+2)}{6} \end{cases} \tag{40}$$

Substituting n_5 into Eq. (40),

$$\begin{cases} a_5 = -J_{\text{dec}}^{\text{real}} \times n_5 \\ f_5 = V_m - J_{\text{dec}}^{\text{real}} \times \dfrac{n_5 \times (n_5+1)}{2} \\ L_5 = V_m \times n_5 - J_{\text{dec}}^{\text{real}} \times \dfrac{n_5 \times (n_5+1) \times (n_5+2)}{6} \end{cases} \tag{41}$$

(2) Compute L_6

Similarly, the deceleration, feedrate, and displacement of segment VI can be described as follows:

$$\begin{cases} a_i = a_5 \\ f_i = f_5 + a_5 \times i \\ L_i = f_5 \times i + a_5 \times \dfrac{i \times (i+1)}{2} \end{cases} \tag{42}$$

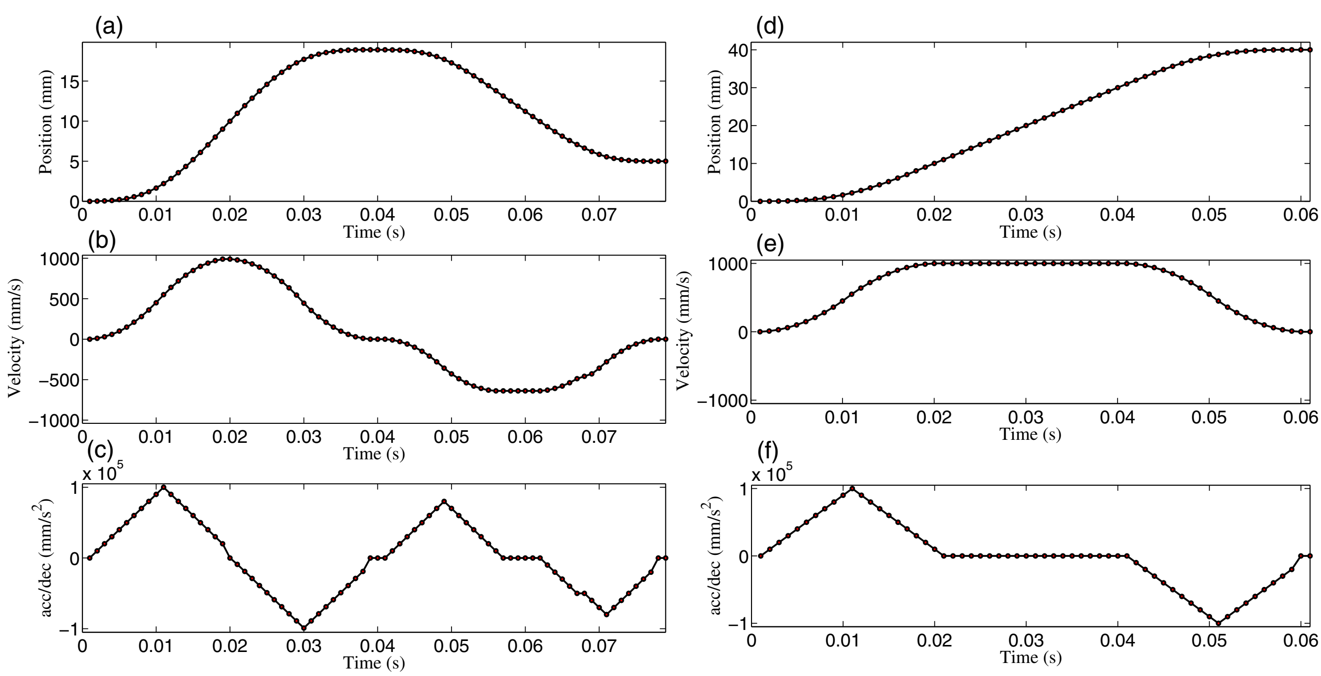

Fig. 11 Simulation on end-point modification algorithm (M_{state} = ACC_PLAN). **a**, **d** Displacement profile; **b**, **e** speed profile; **c**, **f** acceleration/deceleration profile

By substituting n_6, Eq. (42) can be re-expressed as follows:

$$\begin{cases} a_6 = a_5 \\ f_6 = f_5 + a_5 \times n_6 \\ L_6 = f_5 \times n_6 + a_5 \times \dfrac{n_6 \times (n_6+1)}{2} \end{cases} \tag{43}$$

(1) Compute L_7

Similarly, the deceleration, feedrate, and displacement of segment VII can be obtained as follows:

$$\begin{cases} a_i = a_6 + J_{\mathrm{dec}}^{\mathrm{real}} \times i \\ f_i = f_6 + a_6 \times i + J_{\mathrm{dec}}^{\mathrm{real}} \times \dfrac{i \times (i+1)}{2} \\ L_i = f_6 \times i + a_6 \times \dfrac{i \times (i+1)}{2} + J_{\mathrm{dec}}^{\mathrm{real}} \times \dfrac{i \times (i+1) \times (i+2)}{6} \end{cases} \tag{44}$$

Substituting n_7 into Eq. (44),

$$\begin{cases} a_7 = -J_{\mathrm{dec}}^{\mathrm{real}} \\ f_7 = f_6 + a_6 \times n_7 + J_{\mathrm{dec}}^{\mathrm{real}} \times \dfrac{n_7 \times (n_7+1)}{2} \\ L_7 = f_6 \times n_7 + a_6 \times \dfrac{n_7 \times (n_7+1)}{2} + J_{\mathrm{dec}}^{\mathrm{real}} \times \dfrac{n_7 \times (n_7+1) \times (n_7+2)}{6} \end{cases} \tag{45}$$

Therefore, L_{dec} can be computed as follows:

$$\begin{aligned} L_{\mathrm{dec}} = L_5 + L_6 + L_7 = V_m \times n_5 - J_{\mathrm{dec}}^{\mathrm{real}} \\ \times \frac{n_5 \times (n_5+1) \times (n_5+2)}{6} + f_5 \times n_6 + a_5 \\ \times \frac{n_6 \times (n_6+1)}{2} + f_6 \times n_7 + a_6 \\ \times \frac{n_7 \times (n_7+1)}{2} + J_{\mathrm{dec}}^{\mathrm{real}} \times \frac{n_7 \times (n_7+1) \times (n_7+2)}{6} \end{aligned} \tag{46}$$

Considering that Eq. (46) is complex and time-consuming, to satisfy the real-time performance requirement of CNC systems, it should be simplified. The final simplified formula can be expressed as follows:

$$L_{\mathrm{dec}} = (V_m + f_e) \times \left(n_7 + \frac{n_6}{2}\right) + f_e \tag{47}$$

The derivation of formula (47) is given in Appendix 2. The pseudocode of the deceleration displacement computation is given in Algorithm 1.

Algorithm 1 Calculate the deceleration displacement

Input : V_{new}, f_e, D, J_{dec}
Output: $L_{dec}, n_5^{real}, n_6^{real}, n_7^{real}, D^{real}, J_{dec}^{real}$
Procedure
$S_{dec}(V_{cur}, f_e, D, J_{dec}, \&n_5^{real}, \&n_6^{real}, \&n_7^{real}, \&D^{real}, \&J_{dec}^{real})$
  $\Delta V_{dmax} \leftarrow (V_{cur} - f_e);$
  if $V_{dmax} < 0$ **then**
    $n_5^{real} \leftarrow 0, n_6^{real} \leftarrow 0, n_7^{real} \leftarrow 0;$
    $J_{dec}^{real} \leftarrow J_{dec}, L_{dec} \leftarrow V_{cur};$
  else
    $n_5^{real} \leftarrow ceil(\frac{D}{J_{dec}}), \Delta V_{dec} \leftarrow (n_5^{real} * D);$
    if $V_{dec} > V_{dmax}$ **then**
      $n_5^{real} \leftarrow ceil(sqrt(\frac{\Delta V_{dmax}}{J_{dec}})), n_6^{real} \leftarrow 0;$
      $J_{dec}^{real} \leftarrow (\frac{\Delta V_{dmax}}{n_5^{real} * n_5^{real}});$
    else
      $n_d^{real} \leftarrow ceil(\frac{\Delta V_{dmax}}{D}), D^{real} \leftarrow (\frac{\Delta V_{dmax}}{n_d^{real}});$
      $n_5^{real} \leftarrow ceil(\frac{D^{real}}{J_{dec}}), J_{dec}^{real} \leftarrow (\frac{D^{real}}{n_5^{real}});$
      $n_6^{real} \leftarrow (n_d^{real} - n_5^{real});$
    $n_7^{real} \leftarrow (n_5^{real} - 1);$
    $L_{dec} \leftarrow (V_{cur} + f_e) * (\frac{n_6^{real}}{2} + n_5^{real});$
  return L_{dec}

3.4.2 Displacement of segment III

The displacement of segment III can be obtained by integrating Eq. (34) with respect to time:

$$L_i = V_{\mathrm{cur}} \times i + A_{\mathrm{cur}} \times \frac{i \times (i+1)}{2} - J_{\mathrm{acc}}^{\mathrm{real}} \times \frac{i \times (i+1) \times (i+2)}{6} \tag{48}$$

By substituting n_3, Eq. (48) can be rewritten as follows:

$$L_3 = V_{\mathrm{cur}} \times n_3 + A_{\mathrm{cur}} \times \frac{n_3 \times (n_3+1)}{2} - J_{\mathrm{acc}}^{\mathrm{real}} \times \frac{n_3 \times (n_3+1) \times (n_3+2)}{6} \tag{49}$$

Because of the computational complexity of Eq. (49), it should be simplified. The simplified recursive formula can be expressed as follows:

(1) In segment I,

$$L_3 = L_3^{'} + V_{\mathrm{cur}} + 2 \times A_{\mathrm{cur}} \times (n_3 - 1) + J_{\mathrm{acc}}^{\mathrm{real}} \tag{50}$$

(2) In segment II,

$$L_3 = L_3^{'} + A_{\mathrm{cur}} \times n_3 \tag{51}$$

The derivation of formulae (50) and (51) is given in Appendix 3.

3.5 Remaining displacement compensation

From the above calculation, the actual interpolation periods of each segment were obtained except segment IV. The remaining length of the programmed path can be calculated as follows:

$$L_{\mathrm{remain}} = L - L_{\mathrm{cur}} \tag{52}$$

where L_{cur} can be computed as follows:

$$L_{\mathrm{cur}} = \sum_{j=1}^{n_i} V_j \tag{53}$$

where V_j (j = 1, 2, … , n_i) denotes the speed of each interpolation period. Thus, the remaining displacement L_r can be expressed as follows:

$$L_r = L - L_{\mathrm{cur}} - L_{\mathrm{dec}} \tag{54}$$

If $L_r > V_m$, namely, the constant segment IV exists, then L_r will be compensated with the speed of V_m. According to the analysis at the beginning of Sect. 3.4, the following inequality should be satisfied at the final interpolation period of segment IV.

$$0 \le L_r < V_m \tag{55}$$

The Acc/Dec process will become the deceleration zone in the next interpolation period when Eq. (55) is satisfied. Figure 4 shows the schematic of the remaining displacement compensation.

As shown in Fig. 4, L_r can be compensated during the deceleration zone by inserting one more sampling period with the speed of L_r. The criteria for compensation is described as follows:

$$L_r \le V_{\mathrm{cur}} \text{ and } L_r > V_{\mathrm{next}} \tag{56}$$

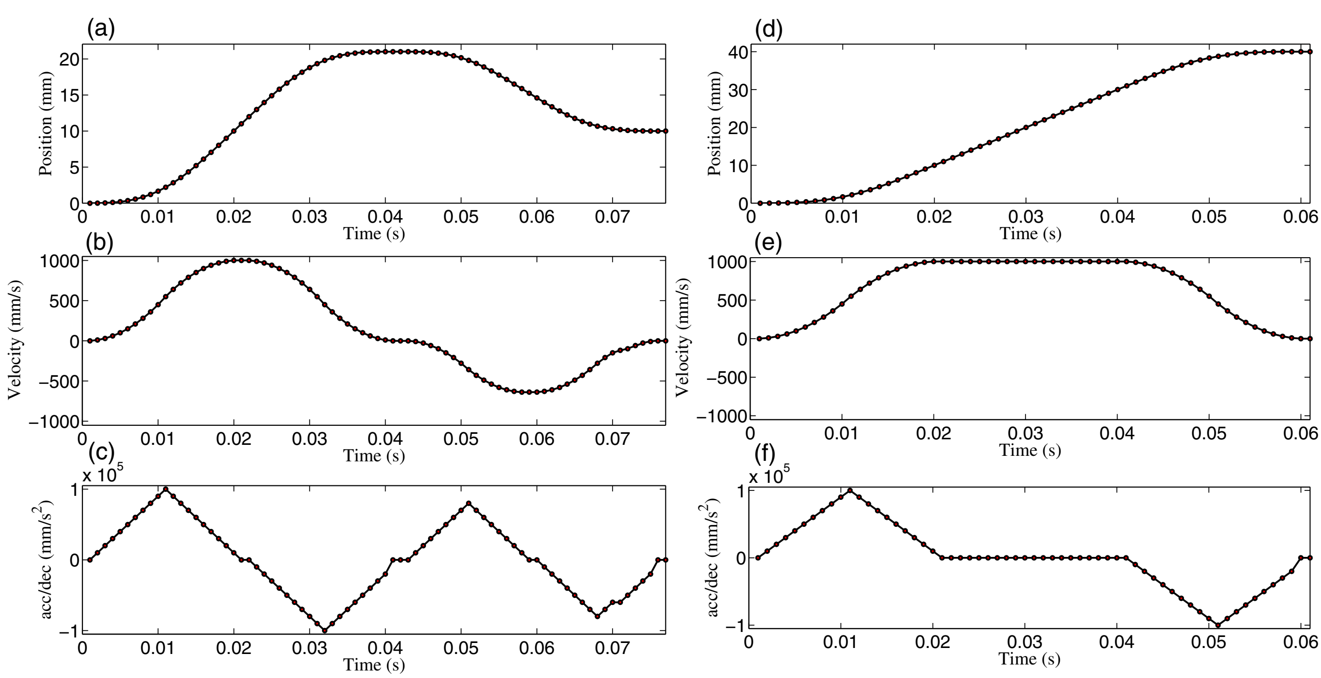

Fig. 12 Simulation on end-point modification algorithm (M_{state} = CON_PLAN). **a**, **d** Displacement profile; **b**, **e** speed profile; **c**, **f** acceleration/deceleration profile

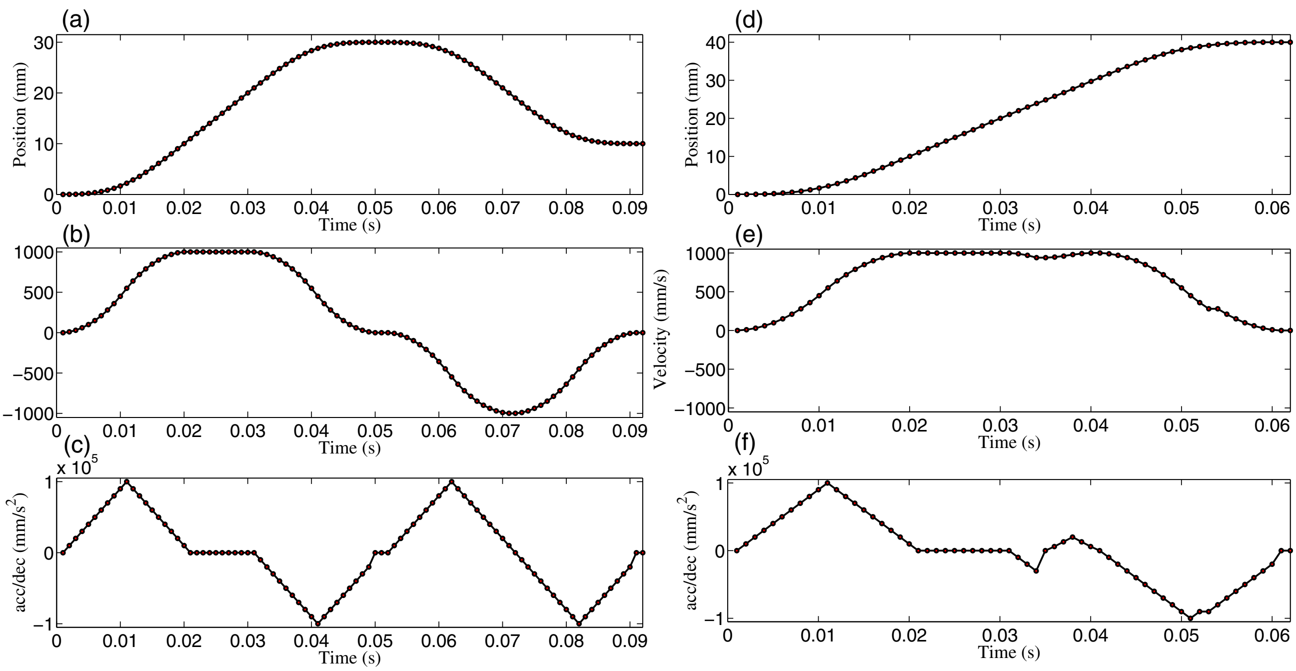

Fig. 13 Simulation on end-point modification algorithm (M_{state} = DEC_PLAN). **a**, **d** Displacement profile; **b**, **e** speed profile; **c**, **f** acceleration/deceleration profile

Table 4 Input parameters of target speed modification algorithm

L (mm)	f_s (mm/s)	f_e (mm/s)	F (mm/s)	$A/D \times 10^3$ (mm/s^2)	$J_{\mathrm{acc}}/J_{\mathrm{dec}} \times 10^6$ (mm/s^3)
3	0	0	50	5/5	1/1

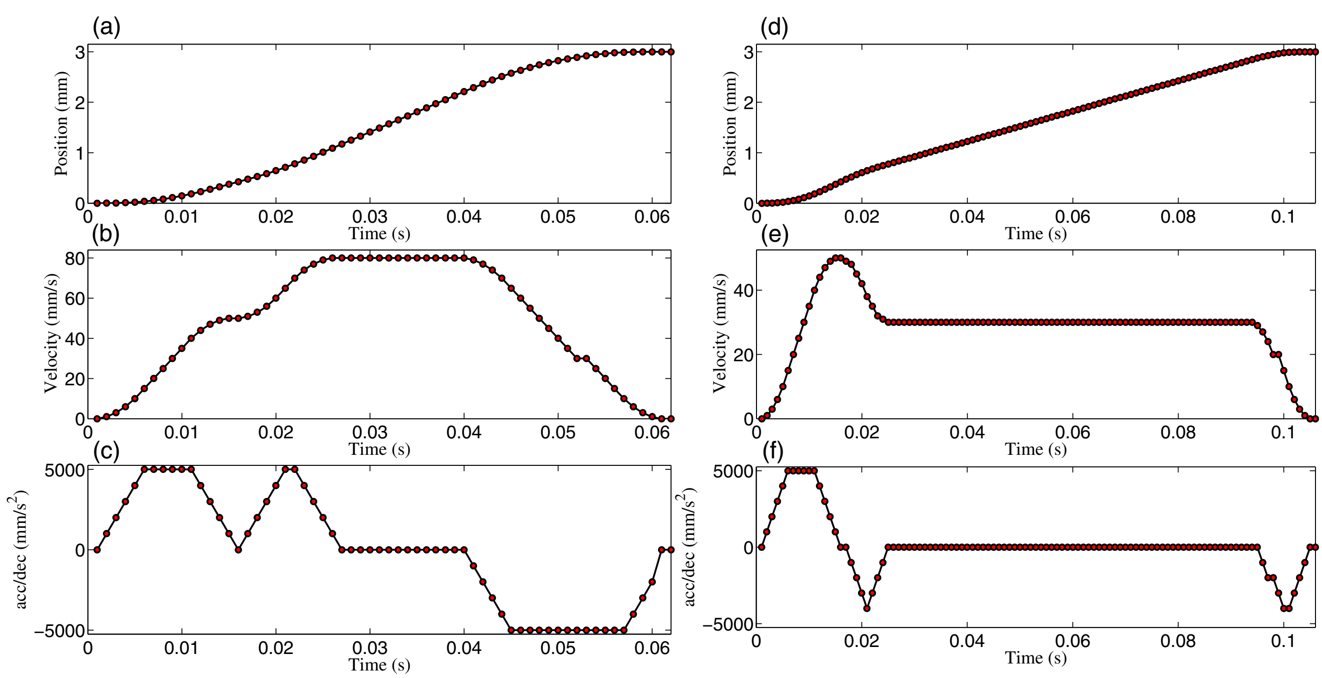

Fig. 14 Simulation on target speed modification algorithm (M_{state} = ACC_PLAN). **a**, **d** Displacement profile; **b**, **e** speed profile; **c**, **f** acceleration/deceleration profile

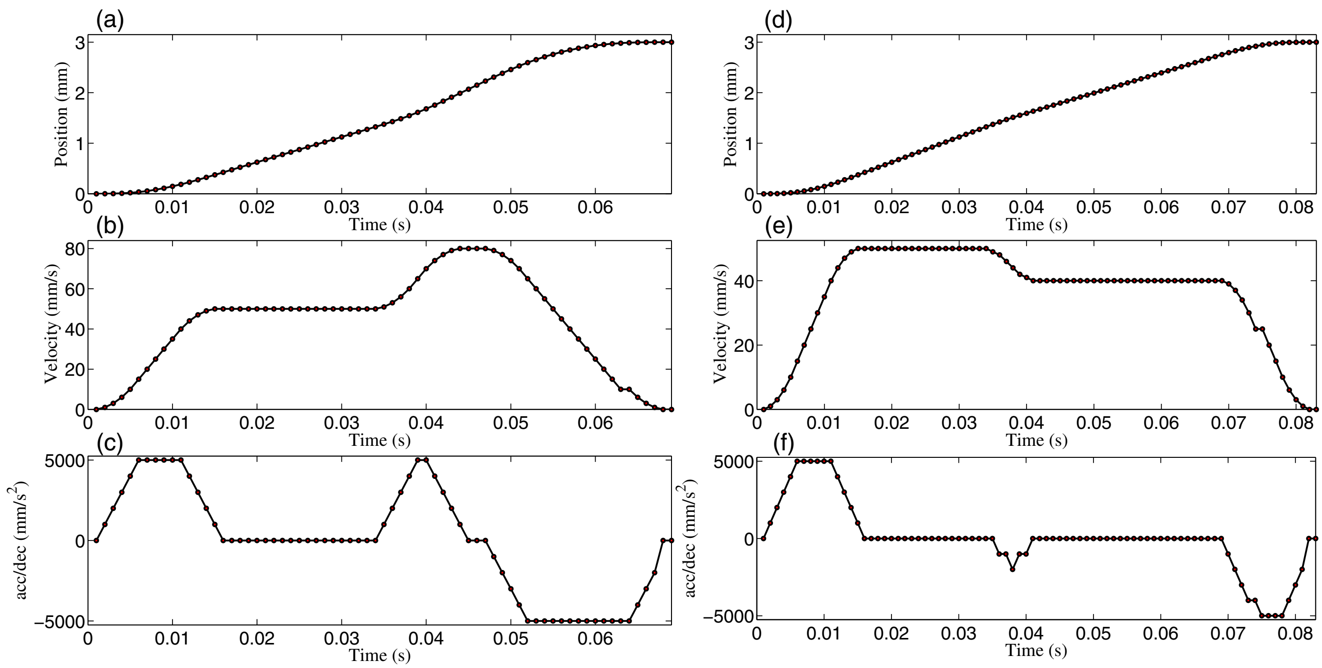

Fig. 15 Simulation on target speed modification algorithm (M_{state} = CON_PLAN). **a**, **d** Displacement profile; **b**, **e** speed profile; **c**, **f** acceleration/deceleration profile

where V_{next} denotes the speed of the next interpolation period. There is a special situation: L_r is smaller than f_e. If so, L_r will be compensated at the last interpolation period with the speed of L_r.

3.6 Velocity planning process implementation

The implementation of velocity planning process is based on the idea of finite state machine (FSM). The state set includes five states: INI_PLAN, ACC_PLAN, CON_PLAN, DEC_PLAN, and END_PLAN. The state transition diagram of FSM is shown in Fig. 5.

The definition and transition conditions among the states are shown in Table 1.

Figure 6 shows the flowchart of the S-curve Acc/Dec velocity planning process.

This can be described as follows:

Step 1: Input initialization parameters;

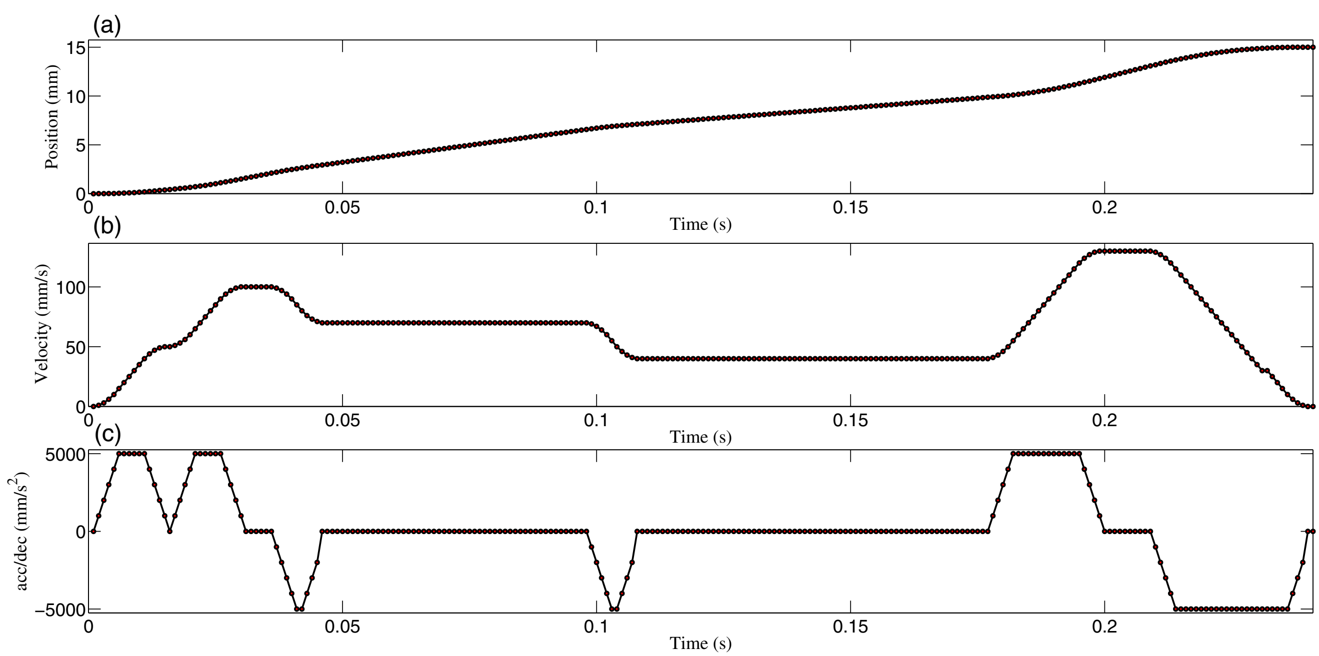

Fig. 16 Simulation of target speed modification algorithm (M_{state} = ACC_PLAN/CON_PLAN). **a** Displacement profile; **b** speed profile; **c** acceleration/deceleration profile

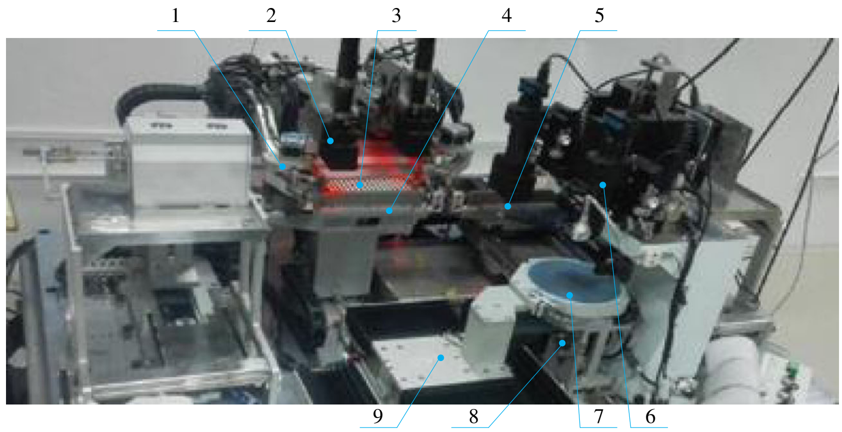

Fig. 17 Photo of die bonder machine. *1*, Paste-dispensing mechanism; *2*, die-bonding optics; *3*, lead frame; *4*, B–C stage; *5*, die-picking and die-bonding mechanism; *6*, die-picking optics; *7*, wafer ring; *8*, push pin mechanism; *9*, X-Y stage

Step 2: Perform the algorithm initialization and stage planning, and then return M_{state};

Step 3: According to the M_{state} returned by step 2, the process will become the corresponding segment by combining with the following conditions:

(a) If M_{state} = ACC_PLAN, compare V_m with F;

a) If $V_m > F$, check n_3;

i. If $n_3 > 0$, go to step 6;

ii. Else go to step 7;

b) Else compare A_{cur} with A;

i. If $A_{\text{cur}} >$ A, then go to step 5;

ii. Else go to step 4;

(b) Else if M_{state} = CON_PLAN, then go to step 7;

(c) Else if M_{state} = DEC_PLAN, check n_5, n_6, and n_7;

a) If $n_5 > 0$, then go to step 8; else check n_6;

b) If $n_6 > 0$, then go to step 9; else check n_7;

c) If $n_7 > 0$, then go to step 10;

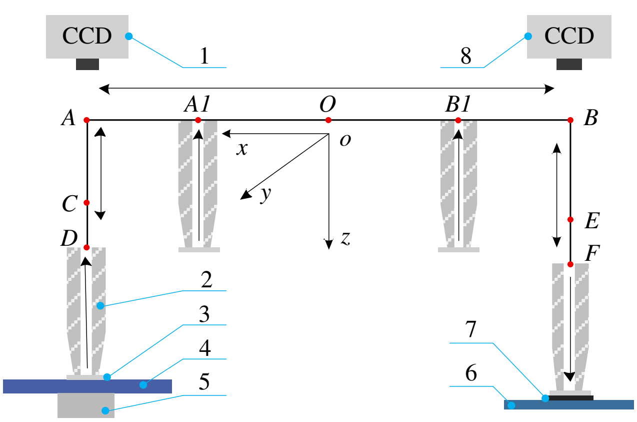

Fig. 18 The execution diagram of the die-picking and die-bonding mechanism. *1*, Die-picking optics; *2*, suction nozzle; *3*, led die; *4*, wafer ring; *5*, push pin mechanism; *6*, lead frame; *7*, silver paste; *8*, die-bonding optics

(d) Else if M_{state} = END_PLAN, then recalculate L_{remain} and compare L_{remain} with f_e;

a) If $L_{\text{remain}} > f_e$, compensate L_{remain} with the speed of f_e. Moreover, let M_{state} unchanged and update V_{cur}, L_{remain}, and L_{cur}.

b) Else compensate the remaining displacement with the speed of L_{remain} and end the algorithm.

Step 4: In segment I, update A_{cur}, V_{cur}, n_1, n_3, L_3, and V_{m}. Then go to step 11;

Step 5: In segment II, update A_{cur}, V_{cur}, n_2, L_3, and V_{m}. Then go to step 11;

Step 6: In segment III, update A_{cur}, V_{cur}, and n_3. Then go to step 11;

Step 7: In segment IV, update J_{acc}, A_{cur}, V_{cur}, n_4, and L_{remain}. Then go to step 12;

Step 8: In segment V, update A_{cur}, V_{cur}, and n_5. Then go to step 13;

Step 9: In segment VI, update A_{cur}, V_{cur}, and n_6. Then go to step 13;

Step 10: In segment VII, update A_{cur}, V_{cur}, and n_7. Then go to step 13;

Step 11: Recalculate L_{dec}, L_{remain}, n_5, n_6, n_7 according to Algorithm 1. Then,

a) If ($V_{\text{m}} \le V_{\text{cur}}$) and ($n_3 = 0$) and ($L_r > V_m$), set ($M_{\text{state}}$ = CON_PLAN), then go to step 14;

b) Else if ($n_3 = 0$) and ($n_5 > 0$), set (M_{state} = DEC_PLAN), then go to step 14;

c) Else if ($n_3 = 0$) and ($n_5 = n_6 = n_7 = 0$), set (M_{state} = END_PLAN), then go to step 14;

d) Else go to step 14;

Step 12: a) If ($0 \le L_r < V_m$) and ($n_5 > 0$), set (M_{state} = DEC_PLAN), then go to step 14;

a) Else if ($0 \le L_r < V_m$) and ($n_5 = n_6 = 0$), set (M_{state} = END_PLAN), then go to step 14;

Step 13: a) If $V_{\text{cur}} > L_r$, compensate L_r with the speed of L_r.

a) Else if ($n_5 = n_6 = n_7 = 0$), set (M_{state} = END_PLAN), then go to step 14;

b) Else go to step 14;

Step 14: Calculate L_{cur}, and then return to step 3.

The flowchart of the initialization algorithm and stage planning is given in Fig. 7.

Moreover, the pseudocode of the initialization algorithm and stage planning is given in Algorithm 2.

4 Extended functionality

In this section, EPMA and TSMA will be introduced.

4.1 EPMA

Figure 8 shows the state transition diagram of EPMA. Once the end-point modification signal (EPMS) is triggered, the

Algorithm 2 Initialization algorithm and stage planning

Input : $L, F, f_s, f_e, A, D, J_{acc}, J_{dec}$ and T_s
Output: M_{state}

$L_{remain} \leftarrow L, L_3 \leftarrow f_s, n_i \leftarrow 0 (i = 1, 2, ..., 7)$;
if $L_{remain} <= f_s$ **then**
  $V_{cur} \leftarrow L_{remain}, L_{cur} \leftarrow V_{cur}, M_{state} \leftarrow END\_PLAN$;
else
  $V_{cur} \leftarrow f_s, V_m \leftarrow V_{cur}, L_{remain} \leftarrow (L_{remain} - V_{cur})$;
  if $F <= f_s$ **then**
    if $L_{remain} > V_{cur}$ **then**
      $L_{dec} \leftarrow S_{dec}(V_{cur}, f_e, D, J_{dec}, \&n_5^{real}, \&n_6^{real}, \&n_7^{real}, \&D^{real}, \&J_{dec}^{real})$;
      $L_r \leftarrow (L_{remain} - L_{dec})$;
      if $L_r < 0$ **then**
        $M_{state} \leftarrow END\_PLAN$;
      else
        $n_5 \leftarrow n_5^{real}, n_6 \leftarrow n_6^{real}, n_7 \leftarrow n_7^{real}$;
        $J_{dec} \leftarrow J_{dec}^{real}, L_{remain} \leftarrow L_r$;
        if $L_{remain} > V_{cur}$ **then**
          $M_{state} \leftarrow CON\_PLAN$;
        else
          $M_{state} \leftarrow DEC\_PLAN$;
    else
      $M_{state} \leftarrow END\_PLAN$;
  else
    $L_{dec} \leftarrow S_{dec}(V_{cur}, f_e, D, J_{dec}, \&n_5^{real}, \&n_6^{real}, \&n_7^{real}, \&D^{real}, \&J_{dec}^{real})$;
    $L_r \leftarrow (L_{remain} - L_{dec})$;
    if $L_r < 0$ **then**
      $M_{state} \leftarrow END\_PLAN$;
    else
      $n_5 \leftarrow n_5^{real}, n_6 \leftarrow n_6^{real}, n_7 \leftarrow n_7^{real}$;
      $J_{dec} \leftarrow J_{dec}^{real}, L_{remain} \leftarrow L_r$;
      $J_{acc}^{real} \leftarrow RecalculateJ_{acc}(\cdot)$;
      $A \leftarrow A^{real}, J_{acc} \leftarrow J_{acc}^{real}, M_{state} \leftarrow ACC\_PLAN$;
return M_{state}

remaining displacement and new end-point should be readjusted as follows:

$$L_r = (L_{\text{remain}} + (L_{\text{new}} \times \text{Dir}-L_s)-L_e) \tag{57}$$

where L_r denotes the remaining displacement between the current position and new end-point. L_{new} is the new end-point. L_s, L_e, and Dir are the starting position, end position and direction of the previous movement, respectively. If $(L_e - L_s > 0)$ is satisfied, Dir = 1; or Dir = −1.

Then, the end position can be updated as follows:

$$L_e \leftarrow \text{fabs}(L_{\text{new}} \times \text{Dir}-L_s) \tag{58}$$

There would be reverse movement compensation when inequality ($L_{\text{remain}} < 0$) is satisfied in the END_PLAN section. The pseudocode of the EPMA is given in Algorithm 3.

4.2 TSMA

Figure 9 shows the state transition diagram of TSMA. Users can change the commanded feedrate on the fly. However, the accuracy of the final position should be guaranteed at the end of Acc/Dec process, or the request should be ignored.

As shown in Fig. 9, when the target speed modification signal (TSMS) is triggered, whether the TSMS will be responded, it depends on the remaining displacement of the tool path calculated using Eq. (54). The pseudocode of the TSMA is given in Algorithm 4.

The pseudocode of the function *RecalculateJ*$_{acc}$ (·) used in Algorithms 3 and 4 is given in Algorithm 5.

5 Simulation and real experiment

5.1 Simulation

To validate the proposed algorithms, simulations on the algorithm and its extended functionality were constructed.

5.1.1 Simulations on S-curve Acc/Dec control algorithm

In the simulation, the algorithm was demonstrated with three groups of input parameters as shown in Table 2. Moreover, the actual achieved speed, actual achieved *A*/*D*, and actual achieved $J_{\text{acc}}/J_{\text{dec}}$ are also shown in Table 2.

Figure 10 shows the simulation results. As described in Table 2, in group 0, the kinematic profiles of an unsymmetrical S-curve are shown in Fig. 10a–d. The actual achieved speed, actual achieved *A*/*D*, and actual achieved $J_{\text{acc}}/J_{\text{dec}}$ were 780.2 mm/s, 19.98/29.93 × 10^3 mm/s^2, and 3.996/4.988 × 10^6 mm/s^3, respectively. Figure 10b, c shows that *F* is not reachable, whereas *A*/*D* is reachable. In group 1, the kinematic profiles are shown in Fig. 10e–h. The actual achieved speed, actual achieved *A*/*D*, and actual achieved $J_{\text{acc}}/J_{\text{dec}}$ were 500 mm/s, 62.38/62.25 × 10^3 mm/s^2, and 7.796/7.781 × 10^6 mm/s^3, respectively. Figure 10f, g shows that *F* is reachable, whereas *A*/*D* is not reachable. In group 2, the kinematic profiles are shown in Fig. 10i–l. The actual achieved speed, actual achieved *A*/*D*, and actual achieved $J_{\text{acc}}/J_{\text{dec}}$ were 500 mm/s, 29.29/29.29 × 10^3 mm/s^2, and 4.882/4.882 × 10^6 mm/s^3, respectively. Figure 10j, k shows that both *F* and *A*/*D* are reachable. Figure 10m–o shows that L_r is compensated at the appropriate position of the deceleration zone by inserting one more sampling period with the speed of L_r.

5.1.2 Simulations on the extended functionality

EPMA Simulation experiments, which are triggered when M_{state} equals to ACC_PLAN, CON_PLAN, and DEC_PLAN, were carried out to validate the EPMA in this section. The input parameters are listed in Table 3.

Figure 11 shows the simulation results when M_{state} equals to ACC_PLAN. As shown in Fig. 11a–c, the kinematic profiles of the end-point varied from 30 to 5 mm. When the end-point changed from 30 to 40 mm, the kinematic profiles are shown in Fig. 11d–f.

Figure 12 shows the simulation results when M_{state} equals to CON_PLAN. Figure 12a–c shows the kinematic profiles when the end-point was varied from 30 to 10 mm. When the end-point changed from 30 to 40 mm, the kinematic profiles are shown in Fig. 12d–f. Figure 13 shows the simulation results when M_{state} equals to DEC_PLAN. Figure 13a–c shows the kinematic profiles when the end-point was varied from 30 to 10 mm. When the end-point changed from 30 to 40 mm, the kinematic profiles are shown in Fig. 13d–f.

TSMA Simulation experiments, which are triggered when M_{state} equals to ACC_PLAN, CON_PLAN, and DEC_PLAN, were carried out to validate the TSMA in this section. The input parameters are listed in Table 4.

The simulation results (M_{state} = ACC_PLAN) are shown in Fig. 14. Figure 14a–c shows the kinematic profiles when the target speed was varied from 50 to 80 mm/s. Figure 14d–f shows the kinematic profiles when the target speed was varied from 50 to 30 mm/s.

The simulation results (M_{state} = CON_PLAN) are shown in Fig. 15. Figure 15a–c illustrates the kinematic profiles when the target speed was varied from 50 to 80 mm/s. Figure 15d–f shows the kinematic profiles when the target speed was varied from 50 to 30 mm/s.

Algorithm 3 End-point modification algorithm

Input : L_{new} and end-point modificatio signal (EPMS)
Output:
if $L_{new} * Dir > 0$ **then**
    $L_e \leftarrow fabs(L_{new} * Dir - L_s)$;
else
    $L_e \leftarrow (L_{new} * Dir - L_s)$;
switch (M_{state}) **do**
    Case ACC_PLAN :
    if $L_r < 0$ **then**
        $L_{dec} \leftarrow S_{dec}(V_{cur}, f_e, D, J_{dec}, \&n_5^{real}, \&n_6^{real}, \&n_7^{real}, \&D^{real}, \&J_{dec}^{real})$;
        $L_{remain} \leftarrow (L_{new} * Dir - L_{cur} - L_{dec})$;
        $M_{state} \leftarrow DEC\_PLAN, n_3 \leftarrow 0$;
    Break;
    Case CON_PLAN :
    if $L_{remain} < L_r \&\& V_{cur} < V_m \&\& L_r > V_{cur}$ **then**
        $L_{remain} \leftarrow (L_r - L_{dec}), M_{state} \leftarrow ACC\_PLAN$;
        $n_1 \leftarrow 0, L_3 \leftarrow (-V_{cur}), J_{acc}^{real} \leftarrow RecalculateJ_{acc}(\cdot)$;
    else
        $L_{remain} \leftarrow L_r$;
    Break;
    Case DEC_PLAN :
    if $L_r > V_{cur}$ **then**
        $L_e \leftarrow fabs(L_{new})$;
        if $V_{cur} < V_m$ **then**
            $L_{dec} \leftarrow S_{dec}(V_{cur}, f_e, D, J_{dec}, \&n_5^{real}, \&n_6^{real}, \&n_7^{real}, \&D^{real}, \&J_{dec}^{real})$;
            $L_r \leftarrow (L_e - L_{cur} - L_{dec})$;
            if $L_r > 0$ **then**
                $n_5 \leftarrow n_5^{real}, n_6 \leftarrow n_6^{real}, n_7 \leftarrow n_7^{real}$;
                $J_{dec} \leftarrow J_{dec}^{real}, L_{remain} \leftarrow L_r$;
                $M_{state} \leftarrow ACC\_PLAN, n_1 \leftarrow 0, L_3 \leftarrow (-V_{cur})$;
                $J_{acc}^{real} \leftarrow RecalculateJ_{acc}(\cdot)$;
                Break;
            $L_{remain} \leftarrow (L_{remain} - V_{cur})$;
            if $L_{remain} > V_{cur}$ **then**
                $M_{state} \leftarrow CON\_PLAN$;
    Break;
    Case END_PLAN :
    $L_r \leftarrow (L_e - L_{cur})$;
    if $L_r < 0$ **then**
        $M_{state} \leftarrow ACC\_PLAN, n_1 \leftarrow 0, L_3 \leftarrow (-V_{cur})$;
        $L_s \leftarrow (L_{cur} * Dir), L_{remain} \leftarrow L_r, L_{cur} \leftarrow 0$;
    Break;
return

Algorithm 4 Target speed modification algorithm

Input : V_{new} and target speed modificatio signal (TSMS)
Output:
if $V_{new} > V_{cur}$ **then**
  if $M_{state} == ACC\_PLAN$ **then**
    $F \leftarrow V_{new}$;
  else if $M_{state} == CON\_PLAN$ **then**
    $L_{dec} \leftarrow S_{dec}(V_{cur}, f_e, D, J_{dec}, \&n_5^{real}, \&n_6^{real}, \&n_7^{real}, \&D^{real}, \&J_{dec}^{real})$;
    $L_r \leftarrow (L - L_{cur} - L_{dec})$;
    if $L_r > V_{cur}$ **then**
      $n_5 \leftarrow n_5^{real}, n_6 \leftarrow n_6^{real}, n_7 \leftarrow n_7^{real}$;
      $J_{dec} \leftarrow J_{dec}^{real}, L_{remain} \leftarrow L_r$;
      $M_{state} \leftarrow CON\_PLAN, F \leftarrow V_{new}$;
      $J_{acc}^{real} \leftarrow RecalculateJ_{acc}(\cdot), L_{cur} \leftarrow (L_{cur} + V_{cur})$;
    else
      Ignore the request;
  else
    Ignore the request;
else
  $L_{dec} \leftarrow S_{dec}(V_{new}, f_e, D, J_{dec}, \&n_5^{real}, \&n_6^{real}, \&n_7^{real}, \&D^{real}, \&J_{dec}^{real})$;
  $L_{dec} \leftarrow (L_{dec} + S_{dec}(V_{cur}, f_e, D, J_{dec}, \&n_5^{real}, \&n_6^{real}, \&n_7^{real}, \&D^{real}, \&J_{dec}^{real})$;
  $L_r \leftarrow (L - L_{cur} - L_{dec})$;
  if $L_r > 0$ **then**
    $n_5 \leftarrow n_5^{real}, n_6 \leftarrow n_6^{real}, n_7 \leftarrow n_7^{real}$;
    $J_{dec} \leftarrow J_{dec}^{real}, L_{remain} \leftarrow (L_r + V_{new})$;
    $M_{state} \leftarrow DEC\_PLAN, F \leftarrow V_{new}, L_{cur} \leftarrow (L_{cur}+V_{cur})$;
  else
    Ignore the request;
return

Algorithm 5 Recalculate A J and $_{acc}$

Input : $V_{cur}, V_{new}, A, J_{acc}$
Output: A^{real}, J_{acc}^{real}
Procedure $RecalculateJ_{acc}(V_{cur}, V_{new}, A, J_{acc}, \&A^{real})$
  $\Delta V_{amax} \leftarrow (V_{new} - V_{cur}),\ n_1^{real} \leftarrow ceil(\frac{A}{J_{acc}})$;
  $\Delta V_{acc} \leftarrow (n_1^{real} * A)$;
  if $\Delta V_{acc} > \Delta V_{amax}$ **then**
    $n_1^{real} \leftarrow ceil(sqrt(\frac{\Delta V_{amax}}{J_{acc}})),\ J_{acc}^{real} \leftarrow (\frac{\Delta V_{amax}}{n_1^{real} * n_1^{real}})$;
  else
    $n_d^{real} \leftarrow ceil(\frac{\Delta V_{amax}}{A}),\ A^{real} \leftarrow (\frac{\Delta V_{amax}}{n_d^{real}})$;
    $n_1^{real} \leftarrow (\frac{A^{real}}{J_{acc}}),\ J_{acc}^{real} \leftarrow (\frac{A^{real}}{n_1^{real}})$;
  return J_{acc}^{real}

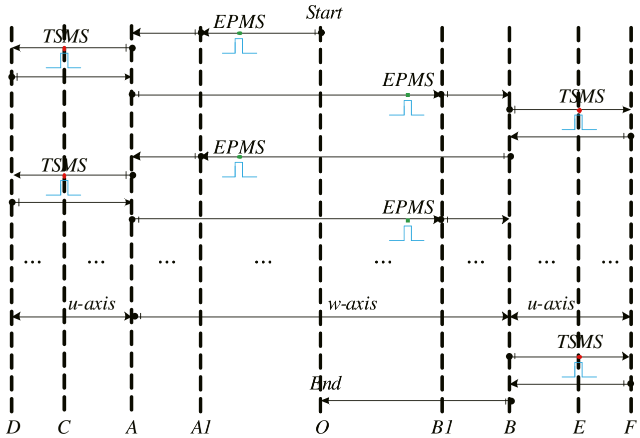

Fig. 19 Timing chart of the die-picking and die-bonding mechanism

Figure 16 shows the simulation on the ability to modify target speed repeatedly. The input parameters are the same as shown in Table 4 except that L was set as 15 mm. Four times requirements were triggered in the simulation. As shown in Fig. 16, the first requirement for the speed changes from 50 to 100 mm/s occurs when M_{state} equals to ACC_PLAN. The other three requirements were triggered when M_{state} equals to CON_PLAN, and the speed changed from 100 to 70, 40, and 130 mm/s, respectively. Finally, the kinematic profiles are shown in Fig. 16a–c.

5.2 Real experiment

The proposed S-curve Acc/Dec control algorithm and its extended functionality were implemented in an embedded motion controller. The validity was verified using a LED die bonder machine as shown in Fig. 17. Taking the die-picking and die-bonding mechanism (DPBM) as an example for testing the algorithms proposed in this paper, its execution diagram is shown in Fig. 18.

As shown in Fig. 18, the DPBM has two degrees of freedom including the rotary motion in the X–Y plane driven by the w-axis and up and down motion in the z-axis direction driven by the u-axis. Points A and B denote the die-picking position (PP) and die-bonding position (BP), respectively. Points A1 and B1 denote the die pre-PP (PPP) and pre-BP (PBP), respectively. Points D and F are the real PP and real BP, respectively. Points C and E are the heights of the PPP and PBP, respectively. O is the origin.

Figure 19 shows the timing chart of the DPBM. As shown in Fig. 19, first, w-axis moves to A1 from O, and EPMS is triggered once the die-picking optics completes the recognition of PP. Therefore, w-axis will move to the new end-point A. Second, u-axis will move to D from A, and TSMS will be triggered when u-axis arrives at C. Then, u-axis will return to PP after picking up the die. Third, w-axis moves to B1 from A, and another EPMS is triggered once the die-bonding optics completes the recognition of BP. Therefore, w-axis moves to the new end-point B. Fourth, u-axis will move to F from B, and another TSMS is triggered when u-axis arrives at E. Then, u-axis will return to B after bonding the die. Finally w-axis returns to O from B and one cycle execution of the DPBM is completed. The real experimental results of the process described above are shown in Figs. 20 and 21. The input

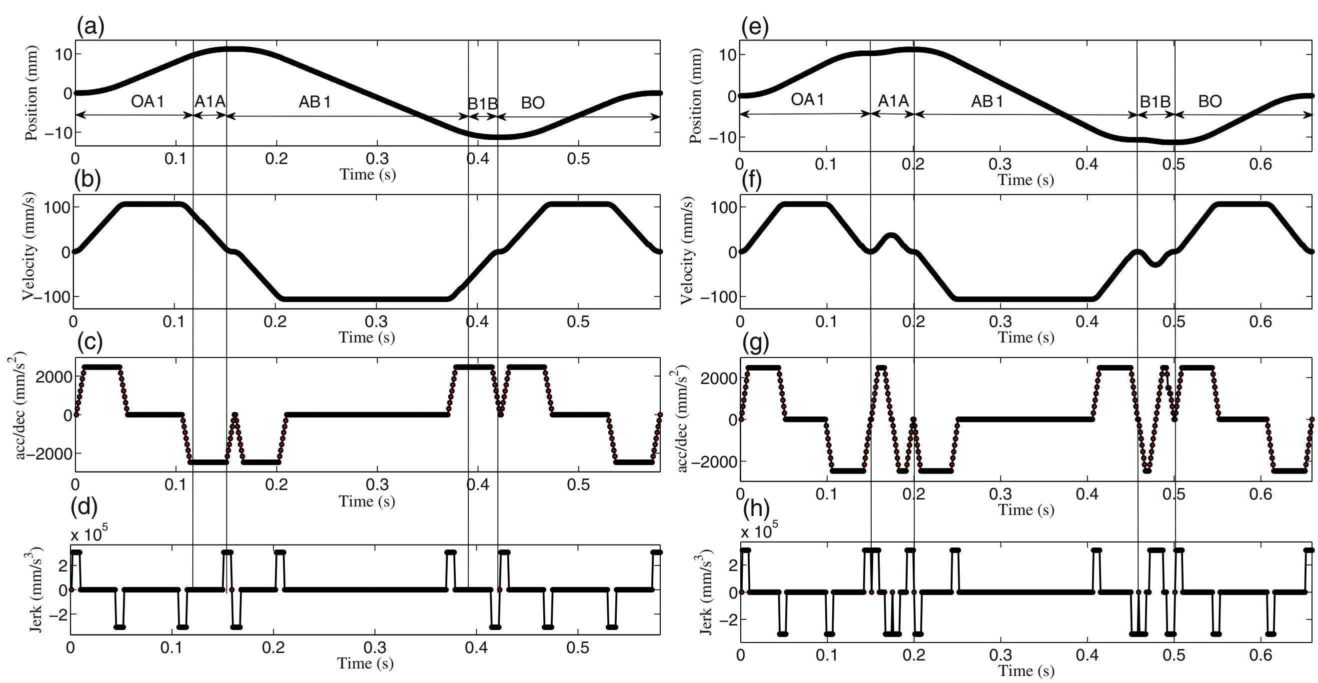

Fig. 20 Comparison of real experimental results in w-axis. **a**, **e** Displacement profile; **b**, **f** speed profile; **c**, **g** acceleration/deceleration profile; **d**, **h** jerk profile

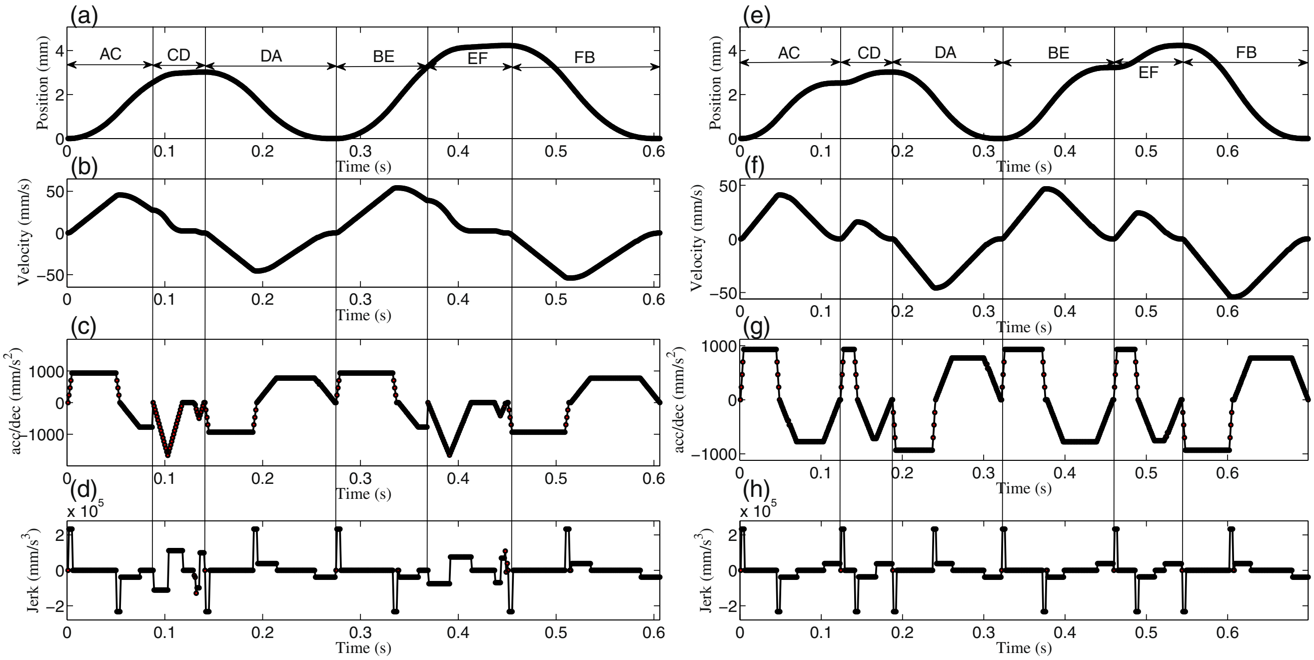

Fig. 21 Comparison of real experimental results in *u*-axis. **a**, **e** displacement profile; **b**, **f** speed profile; **c**, **g** acceleration/deceleration profile; **d**, **h** jerk profile

parameters and displacement of each axis are listed in Tables 5 and 6, respectively.

The kinematic profiles of *w*-axis with EPMA are shown in Fig. 20a–d. Similarly, the kinematic profiles of *w*-axis without EPMA are shown in Fig. 20e–h. Moreover, the kinematic profiles of *u*-axis with TSMA are shown in Fig. 21a–d. The kinematic profiles of *u*-axis without TSMA are shown in Fig. 21e–h. The statistics of processing time in both the situations previously mentioned are shown in Table 7.

As shown in Table 7, the total time was 1358 ms in one cycle without using EPMA and TSMA, whereas it takes only 1187 ms—a savings of 20 % by using EPMA and TSMA. Thus, the statistics indicate that it is capable of speeding up the die-bonding process and increasing the throughput.

6 Conclusion

An unsymmetrical S-curve Acc/Dec control algorithm with confined jerk, acceleration, and speed was constructed in this study. Compared with the conventional S-curve-based Acc/Dec control algorithms, the proposed S-curve Acc/Dec control algorithm derived simpler formulae for calculating the maximal achieved speed, deceleration displacement of the deceleration zone and displacement of constant acceleration segment. Therefore, the computational load and complexity for generating feasible parameters significantly reduced. Moreover, the methods for recalculating the actual interpolation of each segment, actual A/D, and actual J_{acc}/J_{dec} are presented. Furthermore, to satisfy the diversity of consumer requirements, extended functionality of the S-curve Acc/Dec control algorithm is proposed, including the EPMA and TSMA.

The simulation and experimental results show that the S-curve Acc/Dec control algorithm and its extended functionality could be achieved and satisfy the requirements of real-time performance for a control system. Moreover, the entire speed profile and acceleration (deceleration) profiles have high flexibility and continuity. Therefore, the proposed methods satisfy the requirements of speed control for high-speed systems and the feasibility of the proposed algorithms was demonstrated. The experimental case proved that the extended functionality can achieve

Table 5 Input parameters of real experiment

Parameters	f_s (mm/s)	f_e (mm/s)	F (mm/s)	$A/D \times 10^3$ (mm/s^2)	$J_{acc}/J_{dec} \times 10^6$ (mm/s^3)
w-axis	0	0	106.25	2.5/2.5	0.3125/0.3125
u-axis	0	0	62.25	0.9375/0.78125	0.234375/0.039063

Table 6 Displacement of each axis unit (mm)

w-axis				*u*-axis			
A	A1	B	B1	C	D	E	F
11.222	10.320	−11.278	−10.680	2.022	3.022	3.230	4.230

Table 7 Statistics of processing time

Situation	With EPMA			Without EPMA			With TSMA		Without TSMA	
Displacement segments	OA	AB	BO	OA1 + A1A	AB1 + B1B	BO	AD + DA	BF + FB	AC + CD + DA	BE + EF + FB
w-axis										
Time (ms)	158	265	158	199	302	158				
Total time (ms)	581			659						
u-axis										
Time (ms)							274	332	321	378
Total time (ms)							606		699	

~20 % of reduction in the execution time. Thus, the reliability and efficiency of the proposed algorithms were verified.

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Appendix 1

The derivation of the simplified solution for calculating V_m

According to Eq. (35), the achieved speed of the previous interpolation period can be computed as follows:

$$V'_m = V'_{\mathrm{cur}} + A'_{\mathrm{cur}} \times n'_3 - \frac{n'_3 \times (n'_3 + 1)}{2} \times J^{\mathrm{real}}_{\mathrm{acc}} \tag{59}$$

where V'_m, V'_{cur}, A'_{cur} and n'_3 denote the achieved speed, current speed, acceleration, and value of n_3 at the previous interpolation period, respectively.

By subtracting Eq. (35) from Eq. (59) and simplifying, we obtained as follows:

$$\begin{aligned} V_m - V'_m = {} & V_{\mathrm{cur}} - V'_{\mathrm{cur}} + A_{\mathrm{cur}} \times n_3 - A'_{\mathrm{cur}} \\ & \times n'_3 - \left(\frac{n_3 \times (n_3 + 1)}{2} - \frac{n'_3 \times (n'_3 + 1)}{2} \right) \\ & \times J^{\mathrm{real}}_{\mathrm{acc}} \end{aligned} \tag{60}$$

(2) In segment I, we obtained the relationship as follows:

$$\begin{cases} V_{\mathrm{cur}} - V'_{\mathrm{cur}} = A_{\mathrm{cur}} \\ A_{\mathrm{cur}} - A'_{\mathrm{cur}} = J^{\mathrm{real}}_{\mathrm{acc}} \\ n_3 - n'_3 = 1 \end{cases} \tag{61}$$

By substituting Eq. (61), Eq. (60) can be simplified as follows:

$$V_m = V'_m + A_{\mathrm{cur}} \times 2 - J^{\mathrm{real}}_{\mathrm{acc}} \tag{62}$$

(3) Similarly, in segment II, we obtained the relationship as follows:

$$\begin{cases} V_{\mathrm{cur}} - V'_{\mathrm{cur}} = A_{\mathrm{cur}} \\ A_{\mathrm{cur}} = A'_{\mathrm{cur}} \\ n_3 = n'_3 \end{cases} \tag{63}$$

By substituting Eq. (63), Eq. (60) can be simplified as follows:

$$V_m = V'_m + A_{\mathrm{cur}} \tag{64}$$

Appendix 2

The derivation of formula (46)

As shown in Fig. 2b, because the deceleration in the previous n_7 periods of segment V and last n_7 periods of segment VII is symmetrical, therefore, the deceleration zone can be re-planned and divided into four segments, including an accelerated deceleration segment with the previous n_7 periods, a constant deceleration segment with periods from (n_7+1) to (n_7+n_6), a decelerated deceleration segment with periods from $(n_7+n_6+$

1) to $(2\times n_7+n_6)$, and the 4th segment with last one period. L_{dec} can be derived as follows:

According to Eq. (41), the deceleration, feedrate, and displacement at the end of segment V can be computed as follows:

$$\begin{cases} a_{n_5-1} = -J_{\text{dec}}^{\text{real}} \times n_7 \\ f_{n_5-1} = V_m - J_{\text{dec}}^{\text{real}} \times \dfrac{n_7 \times (n_7+1)}{2} \\ L_{n_5-1} = V_m \times n_7 - J_{\text{dec}}^{\text{real}} \times \dfrac{n_7 \times (n_7+1) \times (n_7+2)}{6} \end{cases} \quad (65)$$

where

$$\begin{cases} a_5 - a_{n_5-1} = -J_{\text{dec}}^{\text{real}} \\ f_5 - f_{n_5-1} = a_5 \\ L_5 - L_{n_5-1} = f_5 \end{cases} \quad (66)$$

Similarly, the deceleration, feedrate, and displacement of segment VI can be calculated as follows:

$$\begin{cases} a_i = a_5 = -J_{\text{dec}}^{\text{real}} \times (n_7+1) \\ f_i = f_{n_5-1} + a_5 \times i \\ L_i = f_{n_5-1} \times i + a_5 \times \dfrac{i \times (i+1)}{2} \end{cases} \quad (67)$$

By substituting n_6, Eq. (67) can be re-expressed as follows:

$$\begin{cases} a_{n_6} = a_5 = -J_{\text{dec}}^{\text{real}} \times (n_7+1) \\ f_{n_6} = f_{n_5-1} + a_5 \times n_6 \\ L_{n_6} = f_{n_5-1} \times n_6 + a_5 \times \dfrac{n_6 \times (n_6+1)}{2} \end{cases} \quad (68)$$

Considering f_{n_5-1}, a_{n_5-1}, and Eq. (68), f_{n_6} can be computed as follows:

$$f_{n_6} = V_m - J_{\text{dec}}^{\text{real}} \times \frac{(2\times n_6 + n_7) \times (n_7+1)}{2} \quad (69)$$

Considering a_5 and f_5 in Eq. (41) and f_6 in Eq. (43), we obtained as follows:

$$f_6 - f_{n_6} = a_5 \quad (70)$$

Considering L_6 in Eq. (43), L_{n_6} in Eq. (68), and Eq. (66), we obtained as follows:

$$L_5 + L_6 - L_{n_5-1} - L_{n_6} = f_6 \quad (71)$$

Then, the deceleration, feedrate, and displacement at the end of segment VII can be computed as follows:

$$\begin{cases} a_i = a_{n_6} + J_{\text{dec}}^{\text{real}} \times (i-1) \\ f_i = f_{n_6} + \left(a_{n_6} - J_{\text{dec}}^{\text{real}}\right) \times i + J_{\text{dec}}^{\text{real}} \times \dfrac{i \times (i+1)}{2} \\ L_i = f_{n_6} \times i + \left(a_{n_6} - J_{\text{dec}}^{\text{real}}\right) \times \dfrac{i \times (i+1)}{2} + J_{\text{dec}}^{\text{real}} \times \dfrac{i \times (i+1) \times (i+2)}{6} \end{cases} \quad (72)$$

Substituting n_7 into Eq. (72), L_{n_7} can be calculated as follows:

$$L_{n_7} = f_{n_6} \times n_7 + \left(a_{n_6} - J_{\text{dec}}^{\text{real}}\right) \times \frac{n_7 \times (n_7+1)}{2} + J_{\text{dec}}^{\text{real}} \times \frac{n_7 \times (n_7+1) \times (n_7+2)}{6} \quad (73)$$

Considering Eq. (70), (71), and (73), we obtained as follows:

$$L_5 + L_6 + L_7 - L_{n_5-1} - L_{n_6} - L_{n_7} = f_7 \quad (74)$$

Considering Eqs. (41), (43), and (45) and $f_e = f_7$, we obtained as follows:

$$\begin{aligned} f_e &= f_7 \\ &= f_6 + a_6 \times n_7 + J_{\text{dec}}^{\text{real}} \times \frac{n_7 \times (n_7+1)}{2} \\ &= f_5 + a_5 \times n_6 + a_6 \times n_7 + J_{\text{dec}}^{\text{real}} \times \frac{n_7 \times (n_7+1)}{2} \\ &= V_m - J_{\text{dec}}^{\text{real}} \times \frac{n_5 \times (n_5+1)}{2} - J_{\text{dec}}^{\text{real}} \times n_5 \times n_6 - J_{\text{dec}}^{\text{real}} \times n_5 \times n_7 + J_{\text{dec}}^{\text{real}} \times \frac{n_7 \times (n_7+1)}{2} \\ &= V_m - J_{\text{dec}}^{\text{real}} \times \frac{(n_7+1) \times (n_7+2)}{2} - J_{\text{dec}}^{\text{real}} \times (n_7+1) \times n_6 - J_{\text{dec}}^{\text{real}} \times (n_7+1) \times n_7 + J_{\text{dec}}^{\text{real}} \times \frac{n_7 \times (n_7+1)}{2} \\ &= V_m - J_{\text{dec}}^{\text{real}} \times (n_7+1) \times (n_7+n_6+1) \end{aligned} \quad (75)$$

By substituting f_{n_5-1} and Eq. (75), Eq. (68) can be re-expressed as follows:

$$
\begin{aligned}
L_{n_6} &= f_{n_5-1} \times n_6 + a_5 \times \frac{n_6 \times (n_6+1)}{2} \\
&= \left(V_m - J_{\text{dec}}^{\text{real}} \times \frac{n_7 \times (n_7+1)}{2}\right) \times n_6 - J_{\text{dec}}^{\text{real}} \times (n_7+1) \times \frac{n_6 \times (n_6+1)}{2} \\
&= V_m \times n_6 - J_{\text{dec}}^{\text{real}} \times \frac{n_6 \times (n_7+1) \times (n_7+n_6+1)}{2} = V_m \times n_6 - \frac{n_6 \times (V_m - f_e)}{2} \\
&= \frac{n_6 \times (V_m + f_e)}{2}
\end{aligned}
\tag{76}
$$

Considering L_{n_5-1}, Eqs. (73) and (75), we obtained as follows:

$$
\begin{aligned}
L_{n_5-1} + L_{n_7} &= V_m \times n_7 - J_{\text{dec}}^{\text{real}} \times n_7 \times (n_7+1) \frac{\times (n_7+2)}{6} + f_{n_6} \times n_7 + \left(a_{n_6} - J_{\text{dec}}^{\text{real}}\right) \times \frac{n_7 \times (n_7+1)}{2} + J_{\text{dec}}^{\text{real}} \times n_7 \times (n_7+1) \frac{\times (n_7+2)}{6} \\
&= V_m \times n_7 + f_{n_6} \times n_7 + \left(a_{n_6} - J_{\text{dec}}^{\text{real}}\right) \times \frac{n_7 \times (n_7+1)}{2} \\
&= V_m \times n_7 + \left(V_m - J_{\text{dec}}^{\text{real}} \times \frac{(n_7+1) \times (n_7 + 2 \times n_6)}{2}\right) \\
&\quad \times n_7 - J_{\text{dec}}^{\text{real}} \times \frac{n_7 \times (n_7+1) \times (n_7+2)}{2} \\
&= 2 \times V_m \times n_7 - J_{\text{dec}}^{\text{real}} \times n_7 \times (n_7+1) \times (n_7+n_6+1) \\
&= 2 \times V_m \times n_7 - n_7 \times (V_m - f_e) \\
&= n_7 \times (V_m + f_e)
\end{aligned}
\tag{77}
$$

Considering Eqs. (74), (76), and (77), L_{dec} can be computed as follows:

$$
\begin{aligned}
L_{\text{dec}} = L_5 + L_6 + L_7 \\
&= L_{n_5-1} + L_{n_6} + L_{n_7} + f_7 \\
&= n_7 \times (V_m + f_e) + \frac{n_6 \times (V_m + f_e)}{2} + f_e \\
&= (V_m + f_e) \times \left(n_7 + \frac{n_6}{2}\right) + f_e
\end{aligned}
\tag{78}
$$

Appendix 3

The derivation of formulae (50) and (51)

According to Eq. (49), the value of L_3 in the previous period can be calculated as follows:

$$
\begin{aligned}
L_3' = V_{\text{cur}}' \times n_3' + A_{\text{cur}}' \times \frac{n_3' \times (n_3'+1)}{2} - J_{\text{acc}}^{\text{real}} \\
\times \frac{n_3' \times (n_3'+1) \times (n_3'+2)}{6}
\end{aligned}
\tag{79}
$$

where L_3', V_{cur}', A_{cur}', and n_3' are the achieved displacement, current speed, current acceleration and value of n_3 in the previous period, respectively.

(1) In segment I, the relationship can be described as follows:

$$
\begin{cases}
V_{\text{cur}}' = V_{\text{cur}} - A_{\text{cur}} \\
A_{\text{cur}}' = A_{\text{cur}} - J_{\text{acc}}^{\text{real}} \\
n_3' = n_3 - 1
\end{cases}
\tag{80}
$$

Subtracting the corresponding term in Eq. (49) from Eq. (79) and simplifying, L_3 can be calculated as follows:

$$
L_3 = L_3' + V_{\text{cur}} + 2 \times A_{\text{cur}} \times (n_3 - 1) + J_{\text{acc}}^{\text{real}}
\tag{81}
$$

(2) Similarly, in segment II, the relationship can be expressed as follows:

$$
\begin{cases}
V_{\text{cur}}' = V_{\text{cur}} - A_{\text{cur}} \\
A_{\text{cur}}' = A_{\text{cur}} \\
n_3' = n_3
\end{cases}
\tag{82}
$$

Subtracting the corresponding term in Eq. (49) from (79) and simplifying, L_3 can be obtained as follows:

$$
L_3 = L_3' + A_{\text{cur}} \times n_3
\tag{83}
$$

References

1. Yang Q, Wu Y, Liu D, Chen L, Lou D, Zhai Z, Liu Z (2016) Characteristics of serrated chip formation in high-speed machining of metallic materials. Int J Adv Manuf Technol 86(5):1201–1206. doi:10.1007/s00170-015-8265-x
2. Luo F, Zhou Y, Yin J (2007) A universal velocity profile generation approach for high-speed machining of small line segments with look-ahead. Int J Adv Manuf Technol 35:505–518. doi:10.1007/s00170-006-0735-8
3. Zhang L, You Y, Yang X (2013) A control strategy with motion smoothness and machining precision for multi-axis coordinated motion CNC machine tools. Int J Adv Manuf Technol 64(1):335–348. doi:10.1007/s00170-012-4019-1
4. Tsou C, Chang C, Lai T, Huang C (2013) The implementation and performance evaluation of a silicon-based LED packaging module with lens configuration. Microsyst Technol 19(11):1851–1862. doi:10.1007/s00542-013-1773-4
5. Kwong CK, Chan KY, Wong H (2007) An empirical approach to modelling fluid dispensing for electronic packaging. Int J Adv Manuf Technol 34(1):111–121. doi:10.1007/s00170-006-0552-0
6. Erkorkmaz K, Altintas Y (2001) High speed CNC system design. Part I: jerk limited trajectory generation and quintic spline interpolation. Int J Mach Tools Manuf 41(9):1323–1345. doi:10.1016/S0890-6955(01)00002-5
7. Song F, Yu S, Chen T, Sun L (2015) Research on CNC simulation system with instruction interpretations possessed of wireless communication. J Supercomput 72(7):2703–2719
8. Jeon JW, Ha YY (2000) A generalized approach for the acceleration and deceleration of industrial robots and CNC machine tools. IEEE Trans Ind Electr 47(1):133–139. doi:10.1109/41.824135
9. Jun H, Xiao LJ, Wang YH, Wu ZY (2006) An optimal feedrate model and solution algorithm for a high-speed machine of small

line blocks with look-ahead. Int J Adv Manuf Technol 28(9):930–935. doi:10.1007/s00170-004-1884-2
10. Kang J, Tao T, Mei XS, Wu XT (2003) Exponential acceleration/deceleration algorithm simulation based on digital signal processor. Acta Simulata Systematica Sinica 15(5):678–680. doi:10.16182/j.cnki.joss.2003.05.021
11. Li HK, Li HS, Song LZ, Yin Y, Huang LS, Li W Y (2010) Design of global sliding-mode controlled AC servo controller based on exponential acceleration/deceleration algorithm. IEEE Int. Conf. Mechatronics Autom, ICMA, pp 1507–1511
12. Wang Y, Yang D, Gai R, Wang S, Sun S (2015) Design of trigonometric velocity scheduling algorithm based on pre-interpolation and look-ahead interpolation. Int J Mach Tools Manuf 96:94–105
13. Leng HB, Wu YJ, Pan XH (2008) Research on cubic polynomial acceleration and deceleration control model for high speed NC machining. J Zhejiang Univ Sci A 9(3):358–365. doi:10.1631/jzus.A071351
14. Zheng KJ, Cheng L (2008) Adaptive S-curve acceleration/deceleration control method. In: WCICA 2008 proceedings of 7th World Congress on Intelligent Control and Automation, pp 2752–2756
15. He J, You Y, Chen H, Wang H (2010) A fast nested look-ahead algorithm with S-shape acceleration and deceleration. Acta Aeronautica et Astronautica Sinica 31(4):842–851
16. Chen Y, Wei H, Sun K, Liu M, Wang T (2011) Algorithm for smooth S-curve feedrate profiling generation. Chin J Mech Eng Engl Ed 24(2):237–247
17. Xia Y, Wang W (2012) S-curve acceleration/deceleration algorithm based on reference table. J Appl Sci Eng Technol 4(2):131–134
18. Wang L, Cao J (2013) A look-ahead and adaptive speed control algorithm for high-speed CNC equipment. Int J Adv Manuf Technol 63(5):705–717. doi:10.1007/s00170-012-3924-7
19. Chen Y, Ji X, Tao Y (2013) Look-ahead algorithm with whole S-curve acceleration and deceleration. Adv Mech Eng 2013:1–9. doi:10.1155/2013/974152
20. Lu TC, Chen SL (2016) Genetic algorithm-based S-curve acceleration and deceleration for five-axis machine tools. Int J Adv Manuf Technol 87:1–14. doi:10.1007/s00170-016-8464-0
21. Qiao Z, Wang H, Liu Z, Wang T, Hu M (2015) Nanoscale trajectory planning with flexible Acc/Dec and look-ahead method. Int J Adv Manuf Technol 79(5):1377–1387. doi:10.1007/s00170-015-6911-y
22. Dong J, Wang T, Li B, Ding Y (2014) Smooth feedrate planning for continuous short line tool path with contour error constraint. Int J Mach Tools Manuf 76:1–12
23. Fan W, Gao XS, Yan W, Yuan CM (2012) Interpolation of parametric CNC machining path under confined jounce. Int J Adv Manuf Technol 62(5):719–739. doi:10.1007/s00170-011-3842-0
24. Fan W, Lee CH, Chen JH (2015) A realtime curvature-smooth interpolation scheme and motion planning for CNC machining of short line segments. Int J Mach Tools Manuf 96:27–46